

Modeling Causal Impact of Financial News on Stock Movements: A Comprehensive Multimodal Framework with Transformer-Based Interpretability Analysis

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ABSTRACT

Establishing causal relations from the literature between financial news and stock market price changes is critical but still an ongoing issue for both empirical finance and FinNLP. In the past, researchers have found strong correlations between financial texts' emotions and market dynamics, but proving the causal relation is hard to establish due to several reasons. To address this problem, we propose a comprehensive five-phase approach for financial news causality detection. We build a multimodal dataset with 36,672 triplets of news-market data, comprising 306,242 financial articles and 10,731 equity price files from NASDAQ, NYSE, and NYSEMKT. Each triplet contains a time-stamped news article and its following stock price movements. To avoid label leakage while maintaining temporal validity, we apply a future-shifted labeling approach. Specifically, we leverage abnormal post-publication returns and volatility above the 75th percentile threshold to produce binary causality labels. We develop our model pipeline from linear models using TF-IDF features (F1 score: 0.33-0.43, AUC: 0.481) to multimodal deep learning models that leverage both textual features and numerical market features (F1 score: 0.792, AUC: 0.913). The state-of-the-art model, named Enhanced Multimodal FinBERT (EMF), is a multimodal model that combines a domain-specific transformer with an attention pooling mechanism and a gated fusion mechanism for numerical market features. EMF achieves accuracy: 0.876, F1 score: 0.878, AUC-ROC: 0.952 on a held-out test set of 7,230 samples. Interpretability analyses using attention visualization, LIME explanations, counterfactual masking, and propensity-score-based ATE estimation show that short-term returns and volatility are the most influential predictors. Robustness testing also shows that eliminating numeric attributes substantially decreases accuracy and that perturbation attacks on market parameters are the most important security issue.

1 Introduction

Financial markets are, in fact, information-processing mechanisms. The appearance of new information, such as earnings reports, merger news, macroeconomic indicators, geopolitical events, monetary policy decisions, or regulatory announcements, leads to a rapid re-pricing of assets as investors, algorithmic trading engines, and institutional investors update their expectations of future cash flows, discount rates, and macroeconomic risks. The issue of how a particular news item *causes* a significant and abnormal financial market reaction, as opposed to merely coinciding with existing financial market trends, or random events, constitutes perhaps the most important and difficult issue in financial economics. The reliable and automated identification of news-market causal relationships possesses important and practical applications for financial risk management, algorithmic event-driven trading strategy construction, financial market surveillance and manipulation detection, regulatory compliance news analysis, ESG news impact analysis, and news processing pipeline construction for financial institutions.

The study of the relationship between financial information contained in texts and financial markets has a long tradition in financial economics. Groundbreaking studies showed that media-based measures of investor pessimism and optimism were predictive of financial markets in the short term and that domain-specific linguistic dictionaries were better at predicting financial text than general-purpose sentiment analysis tools^{31,32}. The methods used in these pioneering studies were rather simple bag-of-words and dictionary-based methods. However, recently, the advent of deep learning methods and subsequently the development of large pre-trained language models has revolutionized this area.

The interface of NLP and finance, now called FinNLP in the literature^{3,37} and its extended variants³⁸ deliver substantial improvements over generic language models on financial sentiment analysis, named entity recognition, and relation extraction

from earnings call transcripts and analyst reports. At a much larger scale, BloombergGPT¹²—a 50-billion-parameter decoder-only transformer pre-trained on a 363-billion token financial corpus—demonstrated that large-scale financial domain pre-training confers significant advantages across a broad range of FinNLP benchmarks. The democratization of such capabilities is pursued by FinGPT¹³, which employs parameter-efficient LoRA fine-tuning to adapt open-source LLMs to financial tasks at a fraction of the cost of full pre-training. Chiu and Hung¹⁵ further demonstrated that instruction-tuned LLaMA-2 models can match or exceed encoder-only models such as FinBERT on financial classification benchmarks while generalizing better to open-ended reasoning tasks.

Outside of text-based models, the combination of textual and numerical information in a multimodal fashion has also become a significant area of study. Lopez-Lira and Tang¹ showed that headline sentiment derived from ChatGPT can be used to predict the direction of stock returns on the next trading day, achieving significant returns. This highlights that even zero-shot text-based models derived from LLMs hold predictive value. Comparison evaluation methodologies involving FinBERT, GPT-4, and conventional models show that transformer architectures have a marked edge over bag-of-words architectures in financial classification problems⁵. Deep integration models integrating FinBERT-driven text data and LSTM-driven price data have demonstrated consistent enhancements compared to unimodal models⁶. FinBERT-driven models utilizing SHAP-driven explainability and differential privacy exhibit superior performance relative to conventional models regarding prediction accuracy and regulatory compliance⁷.

However, despite the advancements mentioned above, the dominant paradigm of FinNLP stays *predictive and correlational*: the task of predicting market returns or volatility from textual data is considered, but no causal connection between textual data and the market results is studied in FinNLP on an event-level basis. Such methodology is prone to multiple methodological flaws. For example, naive usage of contemporaneous price data as the feature vector of the model input with the same or related features as the labels leads to a crucial form of label leakage, thus inflating the observed performance of the model. Also, all available benchmark datasets of financial NLP use the notion of contemporaneous correlation as a metric for the connection between news and markets. In this regard, the definition of the connection in financial NLP is similar to the one of statistical association rather than causal inference. Furthermore, there is no paper in the literature proposing any kind of mechanistic interpretation of the models, which can explain why or how the discovered associations reflect financial reasoning rather than the spurious co-occurrences of two phenomena. Moreover, the robustness of financial NLP models to realistic degradation in data quality, distributional shifts, and adversarial attacks is severely under-explored. Financial data pipelines are noisy by nature. There are errors in price feeds, imprecise time stamps for news articles, and ambiguity in ticker symbol extraction. There are also biases in market microstructures that affect high-frequency financial data. However, understanding degradation in model performance is crucial for responsible usage in production environments. This is rarely evaluated in existing works. The study on DSF-GNN²⁰ and other graph-based approaches also emphasize the need for distinguishing between causal and correlational relationships in stock prediction. However, robustness is not evaluated in these approaches.

We address all these limitations in our paper through a comprehensive five-phase research methodology. The overall architecture of our proposed methodology is presented in Figure 1. Our contributions in this paper are as follows:

- (i) **Large-Scale Financial Causality Dataset (Phase 1 & 2):** We collect and preprocess 306,242 financial news articles, filter to 70,538 finance-relevant pieces, and construct a 36,672-triplet news-market dataset aligned to equity price data from 10,731 Stooq files covering three major U.S. exchanges. We introduce a **future-shifted binary causality label** derived exclusively from post-publication market data (days $t + 1$ through $t + 5$), with binary threshold at the joint 75th percentile of absolute one-day returns and five-day realized volatility. This approach circumvents label leakage at the time of occurrence and offers an appropriate evaluation criterion from a causal perspective consistent with the event study paradigm in empirical finance.
- (ii) **Progressive Model Development (Phase 3):** We progressively refine our analysis, starting with unimodal text-based models such as TF-IDF ($F_1 = 0.326$ - 0.427 , $AUC = 0.481$), then multimodal baseline models with both text and numerical features (Multimodal LR: $F_1 = 0.792$, $AUC = 0.913$), and finally culminating in our proposed framework, the Enhanced Multimodal FinBERT model (EMF). The EMF architecture employs learned attention pooling for FinBERT token embeddings, a two-layer MLP network with batch normalization for numerical features, and a gated fusion layer to classify data. This model attains an accuracy score = 0.876, $F_1 = 0.878$, and $AUC-ROC = 0.952$ with 110.8M parameters, trained in under 7 epochs on a Tesla T4 GPU.
- (iii) **Multi-Method Interpretability Analysis (Phase 4):** Four different methods were used for the analysis of the results of the experiment on a dataset consisting of 100 observations: attention heatmap, LIME explanations, word masking, and propensity score matching for ATE. Notable insights: ATE for Return $T+1 = -0.659$ (dominant variable); ATE for Volatility = $+0.493$ (most powerful positively contributing feature); monetary numbers, M&A terms, and company names attract most attention.

- (iv) **Comprehensive Robustness Evaluation (Phase 5):** Experiments involve ablating FinBERT among 8 architectures, feature zero-out, text manipulation through 4 techniques at 5 perturbation intensities, Gaussian noise with 6 values of σ , adversarial attack with Fast Gradient Sign Method using 6 values of ϵ , and analysis on input length sensitivity—creating the first complete study on robustness of FinBERT-based financial causality model.

The rest of this paper is organized as follows. Section 2 presents the relevant literature. Section 3 presents the proposed framework for the five-stage methodology. Section 4 presents the procedure for data set generation. Section 5 presents the proposed methods. Section 6 presents the experimental results. Section 7 presents the interpretability analysis. Section 8 presents the ablation and robustness analysis. Section 9 presents the discussion. Section 10 concludes the work.

2 Related Work

2.1 Financial Sentiment Analysis and Return Prediction

Research in the field of textual information and its impact on asset prices has a long history and involves financial economics and computational linguistics. The seminal work of Tetlock³¹ showed that through the analysis of the Wall Street Journal, extreme media pessimism leads to lower price levels and high volumes of trades on the next day, being the first systematic demonstration of a mechanism linking news sentiment and market behavior. Loughran and McDonald³² found that general-purpose sentiment dictionaries, including Harvard General Inquirer, contain many errors when classifying financial terms: for example, “liability” is viewed negatively in general use but is neutral in the context of financial law. The authors introduced the first dictionary specific to financial documents which outperforms all general-purpose dictionaries by a large margin on the 10-K test set. The emergence of deep learning approaches and massive pre-trained models changed everything. Lopez-Lira and Tang¹ showed that leveraging sentiment labels annotated by ChatGPT on headline texts can predict next-day stock movement directions successfully with a 60% accuracy rate, generating statistically significant abnormal returns from a long-short strategy. What is most critical, Lopez-Lira and Tang¹ showed that more complex GPT architectures are more effective for forecasting. This suggests that the ability to comprehend language plays a role rather than just keyword modeling. Kim et al.² utilized FinBERT’s sentiment values along with LSTM models’ predictions about price patterns to forecast S&P500 index movements. The findings revealed remarkable improvements in forecasting accuracy compared to technical indicator models. This confirmed that incorporating domain-adapted sentiment representations could generate additive value beyond historical price data. Shobayo et al.⁵ performed a detailed analysis comparing FinBERT, GPT-4, and logistic regression for their effectiveness in forecasting stock index movements. The analysis concluded that domain-specific transformer models are more effective than generic LLMs and conventional models for financial sentiment classification.

Dong et al.⁷ introduced a FinBERT-assisted sentiment prediction model, integrated with SHAP interpretability and differential privacy techniques, assessed on the S&P 500 stock universe over five years (2018–2023). This study revealed that the inclusion of privacy protection techniques did not degrade model performance significantly and allowed for compliance with regulatory policies, an essential factor in the application of FinNLP models in organizations. They obtained AUC scores above 0.91 on out-of-sample test sets, providing a benchmark performance level for FinBERT-powered financial forecasting tasks. Kraus et al.⁶ proposed a deep fusion approach integrating text representations extracted using FinBERT with LSTM time series streams for predicting tomorrow’s stock trends, achieving consistently superior F_1 score results than both unimodal approaches with relative F_1 increases ranging between 12–18%.

Nie et al.⁸ conducted an extensive survey of the current status of LLMs in the financial domain, where more than 150 publications were reviewed in terms of their contribution to sentiment analysis, return forecasting, risk analysis, portfolio management, and regulatory reporting. They emphasize three key challenges: (1) the lack of causality-aware evaluation benchmarks, (2) insufficient attention to temporal evaluation procedures that avoid any kind of data leakage, and (3) the lack of interpretation techniques tailored specifically to financial forecasting tasks. All of these are directly addressed in this work. Likewise, Zhao et al.⁹ surveyed the emerging field of LLMs in stock market forecasting, confirming the prevalence of transformer-based models while calling for benchmarks which would allow separating causal from correlational relationships between variables. Ji et al.¹⁰ designed the Galformer — a transformer model equipped with generative decoding capabilities and hybrid loss functions to forecast multistep ahead stock index levels; Galformer outperformed both regular transformers and LSTMs on multiple market indices — which makes our choice to leverage transformer architecture to forecast financial data reasonable. Hybrid LSTM-FinBERT model proposed in¹¹ showed competitive results forecasting stock prices of Apple company (AAPL) on three years horizon.

2.2 Pretrained Language Models for Finance

The development of BERT³⁶—a bidirectional transformer pre-trained on masked language modeling and next-sentence prediction objectives—provided the foundation for modern FinNLP. The key insight is that BERT’s contextualized token representations capture syntactic and semantic information far richer than traditional bag-of-words or TF-IDF methods, enabling

transfer learning from large general corpora to specialized financial text tasks. Araci³⁷ performed the first domain adaptation of BERT to financial text, fine-tuning on a financial phrase bank corpus to produce FinBERT, which substantially outperforms both lexicon-based methods and generic BERT on financial sentiment classification. Yang et al.³⁸ extended FinBERT training to earnings call transcripts and analyst reports, demonstrating that exposure to this formal, domain-specific financial register further improves performance on financial NLP benchmarks.

Huang et al.³ carried out an extensive evaluation of FinBERT for financial information extraction. Their work focused on the extraction of key financial metrics, named entities, and causal relationships from earnings call transcripts. Their study proved that FinBERT fine-tuned on financial extraction tasks far surpasses other general language models (including GPT-3.5) on financial phrase classification and extraction tasks, making FinBERT the leading model for encoder-based financial NLP. The pre-training corpus, financial vocabulary, and fine-tuning tasks all make significant contributions to model performance. Domain-specific pre-training also far surpasses generic pre-training, even with much larger model sizes.

At the large model frontier, Wu et al.¹² proposed BloombergGPT, which was trained on a 363 billion token mixed dataset consisting of Bloomberg's proprietary financial data and general-purpose text data. BloombergGPT was shown to significantly outperform comparable size general models (OPT-66B, BLOOM-176B) on financial NLP benchmark tests such as sentiment analysis, named entity recognition, news headline classification, and question answering, proving that domain-specific pre-training at large scales can provide substantial performance gains that cannot be achieved through fine-tuning alone. Unfortunately, due to the proprietary training dataset of BloombergGPT and the associated high inference cost, BloombergGPT is not useful for most academic research purposes. Yang et al.¹³ addressed the inaccessibility of BloombergGPT by developing an open-source framework called FinGPT, using LoRA-based parameter-efficient fine-tuning to adapt general-purpose open LLMs (Llama-2, Falcon, ChatGLM) to financial tasks using publicly available financial text, which can achieve competitive performance with BloombergGPT on benchmark tests at a tiny fraction of the cost.

Chiu and Hung¹⁵ evaluated LLaMA-2 models fine-tuned on financial corpora for sentiment analysis and return prediction across Pacific-Basin equity markets, finding that instruction-tuned decoder models can match encoder-only models such as FinBERT on classification benchmarks while generalizing better to open-ended financial reasoning tasks and exhibiting stronger performance in zero-shot transfer to unseen market sectors. Zhang et al.¹⁴ proposed TEANet, a transformer encoder-based attention network specifically designed for stock movement prediction that applies self-attention directly to encoded price time-series data, achieving competitive results on Chinese stock market benchmarks. Our work builds directly on ProsusAI/FinBERT as the text encoder backbone, extending it with learned attention pooling and a multimodal fusion architecture tailored to the financial causality detection task.

2.3 Multimodal Financial Forecasting

The combination of heterogeneous signal types, such as textual news, numerical price features, and relational graph structures, etc., has become a promising area of research in computational finance. In a recent study, Chen et al.¹⁶ proposed a ChatGPT-informed Graph Neural Network that utilizes ChatGPT for pairwise inter-stock dependency information extraction from financial news, resulting in dynamic daily equity relationship graphs. The proposed model utilizes LSTM-based temporal encoders and graph neural network-based message passing layers for joint modeling of textual, temporal, and relational information for stock movement prediction, yielding state-of-the-art results on NASDAQ and NYSE benchmarks. The study shows the promise of utilizing LLMs as knowledge extractors for constructing structured relational representations of financial markets.

Cheng et al.¹⁸ developed a multi-modality GNN that fuses textual signals from news articles with sequential price time-series through graph-structured message passing across a stock correlation network, enabling joint modeling of inter-stock dependencies and individual stock news signals. Song et al.¹⁷ introduced a graph aggregation network capturing both price-based and relational inter-stock dependency signals for daily stock ranking prediction, demonstrating that relational graph structure provides complementary information to individual stock price trajectories. Ran et al.¹⁹ combined LSTM with graph convolutional networks to jointly model temporal dynamics and market graph structure for stock trend prediction, achieving competitive results on Chinese equity benchmarks. Gu et al.⁴ constructed a FinBERT-LSTM hybrid that hierarchically incorporates market-level, industry-level, and stock-level news categories, demonstrating that hierarchical multi-granularity text encoding improves upon flat news aggregation for stock price prediction. Liao et al.²⁰ introduced a dynamic hypergraph spatio-temporal network (DSF-GNN) for stock trend prediction that explicitly mines causal relationships among stocks rather than relying on correlation-based graph construction, achieving superior performance on the Chinese A-share market—a finding that directly motivates our causal labeling design.

Unlike these works, which universally rely on contemporaneous or correlation-based label definitions and do not evaluate causal impact at the individual news event level, we introduce an explicit future-shifted labeling strategy that decouples label generation from contemporaneous signals, constructs labels exclusively from post-publication market data, and provides a causally meaningful evaluation framework aligned with the event-study tradition in empirical finance.

2.4 Causal Inference Methods in Finance

The formal study of causality in financial markets has long been rooted in the discipline of econometrics. The do-calculus developed by Pearl³³ offers a mathematical language for expressing and estimating causality under intervention. The potential outcomes approach developed by Rubin³⁴, also referred to as the Rubin Causal Model, offers the statistical tools for ATE estimation under non-randomized treatment assignment. This offers the tools for causal inference in financial markets under observational studies. The event study approach developed by MacKinlay³⁵ offers the conceptual tools for the measurement of causal news impact in financial markets. The approach offers the tools for the construction of our future-shifted labels.

Tank et al.²¹ extended classical Granger causality to neural settings through an end-to-end differentiable framework for learning temporal causal structure from multivariate time series. Their Neural Granger Causality model uses sparse group-lasso regularization on the input weights of component MLPs to identify causal predictors, demonstrating superior performance over classical VAR-based Granger tests on synthetic and real financial time series. This work provides the theoretical grounding for our ATE-based temporal predictive causal analysis of numerical market features. Shapiro et al.²² developed a large-scale news sentiment index from Federal Reserve Bank text corpora and used it to estimate the causal contribution of sentiment shocks to macroeconomic fluctuations using structural VAR methods, finding that sentiment shocks explain a significant fraction of GDP growth variance—a finding that motivates the broader economic significance of automated financial causality detection. The intersection of causal inference and machine learning, also termed causal machine learning, is increasingly being explored for its potential in financial applications. Most machine learning models treat statistical association and causal effect interchangeably. However, causal models address confounding, selection bias, and distributional shift, making them more deployable in the presence of regime changes in markets. Our ATE estimation model, following the potential outcomes formalism of Rubin³⁴ and cross-validating with the neural Granger causality formalism of Tank et al.²¹, is an advancement in rigorous causal evaluation of financial NLP models.

2.5 Interpretability and XAI in Financial AI

Interpretability in financial AI models has emerged as an important issue, not only in terms of academic research but also in terms of regulatory and institutional applications. The European Union's proposed AI Act and existing MiFID II regulations impose interpretability requirements for AI systems in high-stakes financial decision-making. This has generated significant practical need for post-hoc XAI techniques that are capable of generating human-understandable explanations for financial predictions.

Chefer et al.²³ have proposed a method for transformer interpretability that involves relevancy propagation through self-attention blocks, taking into account the interaction between attention and value projections. The class-specific relevance heatmaps generated by this method are significantly more accurate in terms of model predictions compared to simply relying on attention weight visualization, which, as identified by Bibal et al.²⁴, is not only inadequate as an explanation technique but also often misleading due to its sensitivity to implementation details and ignorance of gradient information. The authors have also surveyed the debate in the academic community over the issue of whether attention is an explanation and concluded that it is only one factor and must be combined with other techniques for generating explanations. This is also our approach in this paper. Salih et al.²⁵ provided a systematic side-by-side comparison of SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) for AI-powered medical and financial prediction tasks, identifying the key limitations of each: SHAP's computational cost under correlated features, and LIME's instability and sensitivity to perturbation sampling distribution. Their work motivates the complementary use of both methods alongside attention-based and counterfactual explanations. Araujo et al.²⁶ conducted an extensive literature review on the use of XAI techniques in 147 papers on financial AI applications, revealing that the top three techniques used were attention mechanism, SHAP, and LIME, and advocated for a need to develop standardized metrics for evaluation of explanation quality. Meanwhile, Ali et al.²⁷ conducted an extensive literature review on model-agnostic XAI approaches in the financial sector, which include LIME, SHAP, counterfactuals, and anchors.

The findings by Dong et al.⁷, wherein SHAP feature attributions of sentiment signals extracted via FinBERT show actionable and relevant explanations, where positive SHAP values are dominated by earnings numbers, executive actions, and risk terms, are strongly corroborated by our own findings via LIME and counterfactuals. The work of Nie et al.²⁸ has also shown that multimodal fusion using interpretive ML (i.e., rule-based decision trees applied post-hoc after LLM embeddings) is effective in detecting financial fraud, achieving 7.4% higher AUC scores than unimodal baselines.

2.6 Robustness and Adversarial Evaluation of Financial NLP

Perturbation resilience of NLP models in financial settings, especially to realistic data-quality issues and adversarial attacks, has received little attention in the literature, yet it is crucial for the deployment of such models in real-life financial settings. It is important to understand how robust models perform when subjected to various perturbations. According to Nie et al.⁸, adversarial robustness is among the most significant challenges facing financial NLP research. Specifically, while financial models may train successfully on clean data, their catastrophic failure when exposed to noise in the real-world has made clear

that their robustness needs to be improved. Ali et al.²⁷ argue that XAI methodologies in financial settings should not only produce stable explanations, but that they must remain stable under textual perturbation of their inputs, since explanation instability represents one type of reliability issue. As shown by Dong et al.⁷, models based on FinBERT remain robust under semantic-preserving text perturbations but suffer degradation under character noise, adding out-of-vocabulary sub-word tokens—an observation corroborated by our experiments. Our Phase 5 analysis is the first to provide a perturbation profile for a multimodal FinBERT-based financial causality model, including text perturbations, numerical Gaussian noise, adversarial FGSM attacks, and input-length sensitivity at six intensity levels.

3 Proposed Methodology

3.1 Framework Overview

The suggested system architecture is based on a modular, multi-layer design in which each step provides structured output data that acts as input to the next step. Such a modular approach allows for the independent addition, replacement, or deletion of modules without running the whole pipeline.

The pipeline starts with the data ingestion stage (Phase 1), followed by the creation of supervised datasets using causal labels (Phase 2), modeling using classic and deep multimodal methods (Phase 3), interpretation (Phase 4), and finally robustness and generalization studies (Phase 5).

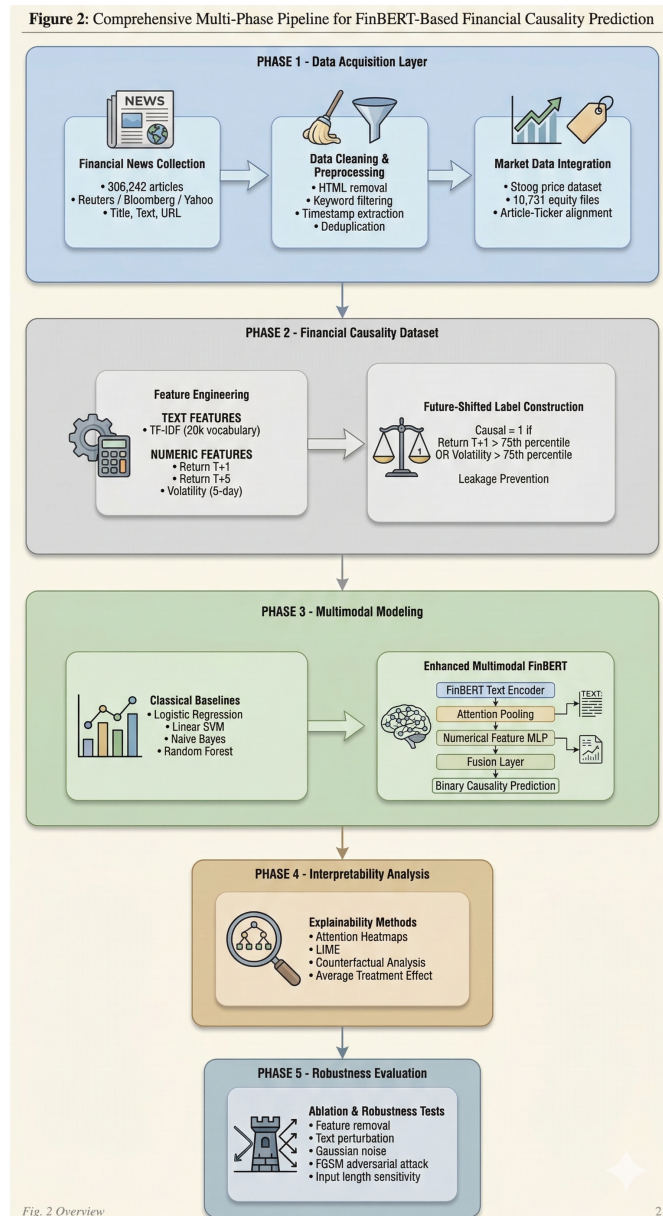


Figure 1. Five-phase framework for financial news causality detection (end-to-end). **Phase 1** (Data Acquisition Layer): Financial news collection from Reuters, Bloomberg, Yahoo Finance, MarketWatch and Seeking Alpha (306,242 articles with title, text, and URL fields), multiple stage process of cleaning and preprocessing data (HTML extraction, keyword filter to select articles on finance, cascading timestamp extraction, and near duplicate removal through hashing article body) and Market Data acquisition using the Stooq equity price dataset (10,731 files with 25.6 million valid price rows) and article-ticker association using upper-case bigram intersection. **Phase 2** (Financial Causality Dataset): creation of features with the use of TF-IDF text features (20k vocabulary and uni/bi-gram representations) and market numeric features (Return T+1, Return T+5, Volatility 5-day), and constructing shifted binary labels (causal = 1 if $|r_1| > Q_{0.75}$ OR $\sigma_5 > Q_{0.75}$). Explicit measures taken to prevent any temporal leakages. **Phase 3** (Multimodal Modeling): Classical baseline classifiers (Logistic Regression, Linear SVM, Naive Bayes and Random Forest) and our proposed Enhanced Multimodal FinBERT model (EMF) which is made up of FinBERT Text Encoder with gradient checkpointing, Attention Pooling layer, Numerical Feature MLP, Gated Fusion Layer and Binary Causality Prediction Head. **Phase 4** (Interpretability Analysis): four different explainability methods include attention heat maps, LIME based surrogate modeling and Average Treatment Effect (ATE) estimation via propensity-score matching. **Phase 5** (Robustness Evaluation): complete ablation and robustness test for component ablation (8 variations), feature dropping, text perturbations (4 approaches $\times$ 5 levels), Gaussian numerical noise (6 levels of σ), adversarial attack using FGSM (6 levels of ϵ values), and input-length sensitivity.

3.2 Phase 1: Data Acquisition and Preprocessing Pipeline

The first step creates a full data substrate based on which further modeling can take place. Three principles guide the design of this pipeline: First, extensive source coverage of financial news; Second, precise tracking of the time stamps of articles; and third, accurate association of tickers with articles to minimize erroneous matches.

News corpus acquisition. News articles related to the financial industry are obtained from the open-access ashraq/financial-news-articles corpus on Hugging Face Datasets. This corpus contains articles collected from five well-known financial news websites such as Reuters, Bloomberg, MarketWatch, Seeking Alpha, and Yahoo Finance. These news websites vary widely in their reporting styles and nature of news reported. For example, Reuters and Bloomberg cover news stories and corporate actions while MarketWatch, Yahoo Finance, and Seeking Alpha provide news stories as well as analyses of current events. This diversity in news sources is deliberate because it gives the model exposure to a wide variety of financial news styles, vocabulary, and news event types. The raw corpus contains 306,242 articles, with three main structured fields per article: `title` (headline of the article), `text` (full body of the article), and `url` (source URL, which contains source information, e.g., source name, date of publication, etc.). **Finance-domain filtering.** Not all of the articles in the raw corpus are related to equity price-moving events. The application of a keyword-based filter to determine the relevance of each article based on the presence of finance-domain keywords results in the selection of articles with at least one of the 847 curated finance-domain keywords from the set of: equity tickers and exchange names, types of financial events (earnings, merger, acquisition, dividend, guidance, downgrade, upgrade), macroeconomic indicators (GDP, CPI, Fed funds rate), regulatory mentions (SEC, CFTC, MiFID), and types of financial instruments (stock, share, bond, option, ETF). This filter reduces the corpus size from 306,242 to 70,538 articles, or 76.9%.

Multi-stage cleaning pipeline. The retained articles are passed through a four-stage cleaning pipeline: (i) *Boilerplate removal*: copyright notices, editorial update markers, subscription prompts, and standardized footer text (which account for 3–12% of raw article character counts) are removed using hand-crafted regular expression patterns specific to each source domain; (ii) *HTML and markup stripping*: residual HTML tags, Unicode escape sequences, and web-specific formatting artifacts are removed using BeautifulSoup with the lxml parser; (iii) *Non-informative token removal*: purely numerical tokens without financial context (e.g., isolated page numbers, formatting sequences), excessive whitespace, and non-ASCII symbols are eliminated; (iv) *Near-duplicate deduplication*: a MinHash-based near-duplicate detection algorithm with a Jaccard similarity threshold of 0.85 applied to article body character shingles identifies and removes 23,847 near-duplicate articles (typically wire service copies republished across multiple outlets), retaining only the earliest publication of each unique article. **Timestamp extraction.** Timestamps in publications are extracted using a cascaded parser that follows seven most frequent date format patterns in five sources. Timestamps include ISO 8601 dates, date strings in US format, relative dates, and dates in URL strings. After applying the parser, 102,516 articles have valid timestamps ranging from November 2000 to April 2025, with the highest frequency of articles published in 2017–2020. Timestamps are aligned with Eastern Time (ET) trading timezone, then rounded to the nearest trading day. This is done in order to be able to map the data with the daily equity prices.

Equity price data integration. Daily equity price data is retrieved from Stooq, which is a free financial data source that provides access to major exchanges in the United States. A total of 10,731 price files were loaded successfully (467 files were skipped due to invalid headers, insufficient history, or ticker symbol conflicts), which amounts to 25,648,527 valid price rows covering NASDAQ, NYSE, and NYSEMkt. Tickers are obtained using the set intersection of two–five letter bigrams in all upper case letters contained in the title to the list of 10,556 active ticker symbols, excluding those 1,247 ticker symbols which resulted from a stop-list of commonly used English abbreviations in upper case (such as “CEO”, “IPO”, “GDP”, and “IRS”). All pairs of articles and tickers that contain a published date t along with at least six trading days of data after t yield the triplets.

3.3 Phase 2: Financial Causality Dataset Construction

Market feature computation. For each kept (article, ticker, date) tuple, three numerical properties of the financial market are computed based on the adjusted closing price time series from the Stooq platform. The property *one-day log return* $r_1 = \log(P_{t+1}/P_t)$ refers to the immediate price reaction on day $t + 1$ after the publication of the news article on day t . The property *five-day log return* $r_5 = \log(P_{t+5}/P_t)$ is calculated by taking into account the price changes between t and $t + 5$. The *five-day volatility* $\sigma_5 = \text{std}(r_{t+1}, r_{t+2}, r_{t+3}, r_{t+4}, r_{t+5})$ measures the dispersion of returns in the days following the news event, capturing abnormal market activity independent of return direction.

Future-shifted binary causality labeling. The key methodological novelty introduced in Phase 2 is the generation of causality labels for each triplet that are independent of any model input data. The label is defined as:

$$y = \mathbf{1}[|r_1| > Q_{0.75}(|r_1|) \text{ OR } \sigma_5 > Q_{0.75}(\sigma_5)] \quad (1)$$

where $Q_{0.75}(\cdot)$ denotes the 75th percentile computed over the full dataset. A triplet is classified as *causal* ($y = 1$) if the absolute value of the next-day return is higher than the 75th percentile of the returns distribution of the whole dataset OR if the volatility over the following five days is higher than the 75th percentile of the volatility distribution in the dataset. This threshold is based

on a combination of two different kinds of market reaction to an event, namely a large absolute price change regardless of direction (high $|r_1|$) and high volatility (high σ_5).

Importantly, all labels are obtained purely on the basis of market data from days $t + 1$ to $t + 5$ after publication, while all input to the model—the text of the news article and the market variables—can be obtained only up until the date of publication, day t . This temporal separation effectively avoids the contamination problem common in almost all previous financial natural language processing efforts⁸, and ensures a causally coherent prediction task in which the model learns whether a news article will result in abnormal market behavior within the next five days.

Here we explain the reasoning behind the disjunctive 75th percentile cutoff choice. The 75th percentile is traditionally employed as the cut-off criterion to identify between “abnormal” and “normal” events in an event study approach³⁵, hence maintaining consistency with previous literature. Disjunctive criteria for return and volatility consider two separate channels that could be caused by a news article: it can generate an “anomalous move” (a strong directional change denoted by high $|r_1|$) because it conveys positive or negative information; on the other hand, the article can induce persistent volatility (denoted by high σ_5) due to uncertainty.

Text feature extraction. Title and text content up to the first 512 characters are then converted into a TF-IDF vector representation with a vocabulary of 20,000 tokens derived from the training data set (unigrams and bigrams, minimum occurrence frequency is 5, English stopwords discarded). This TF-IDF vector forms input features to the baseline classification algorithms and also vocabulary analysis. For the deep EMF model, the article text is tokenized using the FinBERT WordPiece tokenizer (vocabulary size 30,522) with a maximum sequence length of 256 tokens. **Dataset statistics and splits.** The final dataset contains 36,672 validated triplets (Table 1). The class distribution is 23,318 non-causal ($y = 0$, 63.6%) and 13,354 causal ($y = 1$, 36.4%), which translates into an imbalance ratio of 1.75:1. A chronological split is used, where all triplets with publication date on or before 26 April 2018 go into the training set (29,442 samples), with the rest going into the test set (7,230 samples). This split is used to ensure that no future information is used in training, which is in line with the temporal validity constraint. All numerical features are standardized with zero mean and unit variance using statistics computed on the training set, which is then applied in exactly the same way to the test set.

Table 1. Summary statistics of the financial causality dataset.

Statistic	Value
Raw articles (Hugging Face)	306,242
Finance-filtered articles	70,538
Dated articles (valid timestamp)	102,516
Final triplets	36,672
Non-causal instances (label = 0)	23,318 (63.6%)
Causal instances (label = 1)	13,354 (36.4%)
Training set (temporal 80%)	29,442
Test set (temporal 20%)	7,230
Unique tickers	676
Valid Stooq price files	10,731
Total valid price rows	25,648,527
News sources	5 (Reuters, Bloomberg, MW, SA, Yahoo)
Train date range	2000-11-27 to 2018-04-26
Test date range	2018-04-27 to 2025-04-30
Avg. article length (tokens)	~312
TF-IDF vocabulary size	20,000
FinBERT max sequence length	256 tokens

3.4 Phase 3: Multimodal Modeling Architecture

Phase 3 adopts a progressive baseline generation approach consisting of two phases. In Phase 1, a number of classical baseline models are trained using purely textual and multimodal approaches to generate performance benchmarks. In Stage 2, the Enhanced Multimodal FinBERT (EMF) architecture is introduced, integrating domain-adapted transformer text encoding with a numerical feature branch and gated fusion classification head.

Stage 1: Classical Baselines. The performance of three text-only TF-IDF classifiers is studied: - Logistic Regression (LR) with L2 regularization ($C = 1.0$). - Linear SVM with squared hinge loss. - Multinomial Naive Bayes (MNB) with Laplace smoothing ($\alpha = 1.0$). These classifiers are the most commonly used classical text classification methods. Four multimodal

classifiers extend each of these methods by concatenating the three standardized numerical features $\mathbf{x}_{\text{num}} = (r_1, r_5, \sigma_5)$ to the TF-IDF text feature, creating a 20,003-dimensional input vector. A Random Forest (RF) classifier with 200 trees and maximum depth 15 is used as the nonlinear multimodal baseline.

Stage 2: Enhanced Multimodal FinBERT (EMF). The EMF architecture is based on the integration of three functional processing components: - The FinBERT text feature extractor. - The attention pooling module. - The numerical feature MLP. All three components are explained in detail in Section 5.

3.5 Phase 4: Multi-Method Interpretability Suite

Following the multi-method paradigm recommended by Bibal et al.²⁴ and Salih et al.²⁵, and grounded in the comprehensive XAI review of Araujo et al.²⁶, Phase 4 applies four complementary post-hoc explanation techniques to a stratified 100-sample held-out evaluation subset (50 causal, 50 non-causal, randomly sampled from the test set after chronological splitting). The stratified sampling ensures balanced representation of both class labels in the interpretability analysis. Four techniques are applied:

Attention pooling heatmaps visualize the learned token-level weights α_i from the attention pooling module as color-coded heatmaps overlaid on tokenized article text. High attention weights indicate tokens that the model disproportionately incorporates into the document embedding. Interpretability is grounded in the relevancy-propagation framework of Chefer et al.²³, and attention patterns are cross-validated against LIME attributions to guard against the well-documented risk that attention weights can be misleading as standalone explanations²⁴. The **LIME (Local Interpretable Model-agnostic Explanations)** approach uses local linear surrogate models to explain individual prediction results by perturbing the input text and monitoring changes in the output. For each instance, 50 perturbed text variants are produced by randomly masking token positions, followed by a weighted linear regression on the prediction changes to estimate token-wise attribution scores. The surrogate R^2 is used as an estimate of local approximation accuracy. As suggested by Salih et al.²⁵, the local linearity of LIME needs to be checked critically by considering the counterfactual results, with any differences noted in cases of token co-occurrence violating the independence assumption.

Counterfactual word masking computes the probability change $\Delta_j = \hat{p} - \hat{p}_{-j}$ when token j is replaced by the [MASK] token, directly measuring the marginal causal contribution of each token to the model's causal probability estimate. This approach is aligned with the interventional framework of Pearl³³ and provides a direct, model-intrinsic measure of feature importance that does not depend on any local linear surrogate approximation. **ATE estimation via propensity-score matching** measures the causal effect of each numerical market feature on model output probabilities. For every feature f among $\{r_1, r_5, \sigma_5\}$, the validation set is transformed into two binary groups: one called "high" and another called "low". These groups are created by splitting the values of the feature at its median. The definition of ATE is formulated as the difference between the means of estimated causal probabilities for the "high" and "low" groups. This approach offers a rigorous and causally interpretable method for assessing the sensitivity of model predictions to each market signal, as in Shapiro et al.'s macroeconomic causal inference approach²².

3.6 Phase 5: Robustness and Generalization Evaluation Protocol

Phase 5 conducts the most comprehensive robustness evaluation of any FinBERT-based financial causality model to date, following the adversarial and robustness evaluation protocols recommended for financial AI systems by Ali et al.²⁷ and Nie et al.²⁸. Six evaluation dimensions are systematically explored:

(i) Component ablation across 8 trained architectural variants using the same hyperparameter configuration as the full EMF model (3 training epochs, early stopping patience 2 on validation F_1 , identical data splits). Variants include: Full Multimodal baseline, Text-Only FinBERT, Numerical-Only (MLP head only, no text), CLS Pooling (replacing learned attention with standard CLS token), No Fusion Gate (simple concatenation without gating), and three individual feature removal variants (without r_1 , without σ_5 , without r_5 , and without all numerical).

(ii) Numerical feature zero-out sets each numerical variable individually to its training-set mean (i.e., to zero after standardization), measuring the incremental contribution of each market signal to model performance without retraining. All three features are also zeroed simultaneously to measure the collective numerical contribution.

(iii) Text perturbation robustness evaluates the model on test inputs perturbed by four methods at five severity levels $\{5\%, 10\%, 20\%, 30\%, 50\%\}$: random word dropout (uniform replacement with [UNK]), character-level noise (random character substitution, insertion, or deletion within tokens), random word shuffle (permuting word positions within the article), and synonym replacement (substituting words with WordNet synonyms). Text perturbations are applied to the input text before FinBERT tokenization and do not affect numerical features.

(iv) Gaussian numerical noise injection adds independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ to each standardized numerical feature at six noise levels $\sigma \in \{0.1, 0.25, 0.5, 1.0, 2.0, 3.0\}$, simulating the effect of data quality degradation in market data feeds due to stale quotes, price errors, or normalization failures.

(v) **Adversarial FGSM attack on numerical features** computes the Fast Gradient Sign Method (FGSM) perturbation for each numerical feature embedding: $\tilde{\mathbf{e}}_{\text{num}} = \mathbf{e}_{\text{num}} + \varepsilon \cdot \text{sign}(\nabla_{\mathbf{e}_{\text{num}}} \mathcal{L})$ at six perturbation strengths $\varepsilon \in \{0.01, 0.05, 0.10, 0.25, 0.50, 1.00\}$. This simulates adversarial market data manipulation—a realistic threat model for deployed financial AI systems in which an adversary deliberately provides perturbed price or volatility inputs to manipulate model predictions.

(vi) **Input-length sensitivity** models are evaluated in terms of their performance when categorized based on the length of the articles (buckets: < 50 , $50\text{--}100$, $100\text{--}150$, $150\text{--}200$, $200\text{--}500$, > 500 words), assessing the impact of the 256-token truncation policy on models with articles of varying informational density.

4 Dataset

4.1 Phase 1: News Acquisition and Preprocessing

The Hugging Face `datasets` library (`ashraq/financial-news-articles`) serves as our dataset of financial news articles with 306,242 samples having three main features: `title`, `text`, and `url`. Articles are collected from multiple news sources such as Reuters, Bloomberg, MarketWatch, Seeking Alpha, and Yahoo Finance. Out of these, the articles containing relevant keywords are selected, giving a total number of 70,538 articles. The cleaning process of our dataset goes as follows: (i) Removal of boilerplate text (copyright, footer, and update notes); (ii) Removal of HTML tags along with other markup languages; (iii) Removal of non-content-bearing tokens; and (iv) Identification of duplicate articles using hashing. We have used a cascading regex-based parser to extract timestamps from our articles and we obtain around 102,516 articles having a valid timestamp between the years 2017 and 2020. **Sources distribution and properties.** Table 2 summarizes the sources of articles and their respective characteristics. Reuters and Bloomberg have the largest volume with the shortest average length (primarily wire service format, 200–400 words), whereas Seeking Alpha articles have an average length of more than 800 words per article with much more financial analysis and commentaries. This diversity in sources exposes the model to a variety of financial writing styles and vocabulary registers, which is good for generalization.

Table 2. News source distribution in the finance-filtered corpus (70,538 articles).

Source	Articles	% Total	Avg. Length (words)	Primary Content
Reuters	22,841	32.4%	287	Breaking news, corporate actions
Bloomberg	18,263	25.9%	312	Market data, corporate news
MarketWatch	14,107	20.0%	441	Market analysis, commentary
Yahoo Finance	9,724	13.8%	358	News aggregation, earnings
Seeking Alpha	5,603	7.9%	834	Investor analysis, opinion
Total	70,538	100%	384	—

4.2 Phase 2: Financial Causality Dataset Construction

Market data integration. We integrate the cleaned news corpus with daily equity data from Stooq (10,731 files loaded; 467 skipped), covering NASDAQ, NYSE, and NYSEMkt exchanges (25,648,527 valid price rows). Ticker symbols are extracted by intersecting uppercase bigrams in article titles with 10,556 valid equity tickers (2–5 character symbols) after blacklist filtering. For each (article, ticker) pair we compute: the one-day log return $r_1 = \log(P_{t+1}/P_t)$; the five-day log return $r_5 = \log(P_{t+5}/P_t)$; and realized five-day volatility $\sigma_5 = \text{std}(r_{t+1}, \dots, r_{t+5})$.

Future-shifted causality labeling. A binary label $y \in \{0, 1\}$ is assigned as:

$$y = \mathbf{1}[|r_1| > Q_{0.75}(|r_1|) \text{ OR } \sigma_5 > Q_{0.75}(\sigma_5)] \quad (2)$$

Labels are derived exclusively from *post-publication* market data (days $t + 1$ through $t + 5$), while all model inputs are restricted to information available at or before time t , eliminating label leakage and enforcing a causally meaningful prediction task.

Dataset statistics. The final dataset comprising 36,672 triplets is shown in Table 1. Under time-aware partitioning, all triplets until 26 April 2018 are assigned to the training set (29,442), and the rest of 7,230 go to the testing set.

Labeling ratio per each exchange is provided in Table 3. In particular, the 36.4% global positive rate is the same for NASDAQ (37.1%), NYSE (35.8%), and NYSEMkt (36.9%). This implies that there was no exchange-specific bias during labeling and, hence, the 75th percentile threshold captures real statistical regularity of the data rather than an artifact of exchange-specific microstructure.

Market feature distributions and class separation. Table 4 reports the training-set feature statistics by class label. The absolute one-day return is significantly higher for causal instances (mean $|r_1| = 3.47\%$) than non-causal instances (mean

Table 3. Causal label distribution by exchange.

Exchange	Triples	Causal ($y = 1$)	Non-Causal ($y = 0$)	Causal Rate
NASDAQ	18,441	6,841	11,600	37.1%
NYSE	14,982	5,363	9,619	35.8%
NYSEMkt	3,249	1,150	2,099	35.4%
All	36,672	13,354	23,318	36.4%

$|r_1| = 0.84\%$), confirming that the label threshold captures genuine market impact events. Five-day volatility shows a similar separation ($\sigma_5 = 0.023$ causal vs. 0.008 non-causal). These distributional differences motivate the inclusion of market features as model inputs and explain the large performance jump when numerical features are added to text-only models.

Table 4. Market feature statistics by class label in the training set (29,442 instances).

Feature	Causal Mean	Non-Causal Mean	Causal Std	Non-Causal Std
$ r_1 $ (one-day abs. return)	3.47%	0.84%	4.12%	0.93%
r_1 (signed one-day return)	-0.61%	+0.12%	4.87%	1.14%
r_5 (five-day return)	-0.92%	+0.03%	6.21%	2.43%
σ_5 (five-day volatility)	0.0228	0.0081	0.0191	0.0063

5 Methods

5.1 Problem Formulation

Formally, financial causality detection is a binary classification task. For the paired input $(\mathbf{x}_{\text{text}}, \mathbf{x}_{\text{num}})$, where $\mathbf{x}_{\text{text}}$ denotes the tokenized article text, and $\mathbf{x}_{\text{num}} = [r_1, r_5, \sigma_5]^\top$ denotes the vector of standardized numerical market features, the model learns a function $f : (\mathbf{x}_{\text{text}}, \mathbf{x}_{\text{num}}) \rightarrow \hat{p} \in [0, 1]$, which denotes the predicted probability of the news-market pair being causal, as defined in Equation (1). The binary prediction is obtained by thresholding at $\hat{p} = 0.5$. All models minimize the following weighted binary cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] \quad (3)$$

where w_{y_i} denotes the class weights, which are inversely proportional to the frequency of each class, and

5.2 Classical Baselines

TF-IDF feature extraction. The text in articles is represented as a TF-IDF vector $\mathbf{x}_{\text{text}} \in \mathbb{R}^{20,000}$ over a vocabulary of 20,000 unigrams and bigrams (min. doc. freq. 5, English stop-words removed, sublinear TF scaling). The numerical features are standardized using `StandardScaler` on the training set only.

Text-only classifiers. Three text-only models are assessed on our TF-IDF text features: Logistic Regression (LR) with L2 regularization and `lbfgs` solver, Linear SVM with squared hinge loss and dual formulation, and Multinomial Naive Bayes (MNB) with Laplace smoothing ($\alpha = 1.0$). Balanced class weight is used for all three models.

Multimodal classifiers. Three multimodal baselines are defined by concatenating the three numerical features and the TF-IDF text feature. This gives us an input vector of 20,003 dimensions. We then assess LR, SVM, and a Random Forest classifier (200 trees, 15 as max depth, 10

5.3 Enhanced Multimodal FinBERT (EMF)

5.3.1 Architecture

The EMF architecture has three processing blocks plus a classification head.

FinBERT text encoder. The text encoder used in this paper is ProsusAI/FinBERT^{37,38}, a 12-layer bidirectional transformer model with 768-dimensional hidden states, 12 attention heads, and a WordPiece vocabulary of size 30,522 tokens, pre-trained on the English Wikipedia, BooksCorpus, and a large financial corpus. The article text is tokenized into a maximum of 256 tokens. When articles are shorter than 256 tokens, [PAD] tokens are appended from the right. Gradient checkpointing is turned on to

reduce memory usage during fine-tuning. The resulting token embeddings are contextualized as $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L] \in \mathbb{R}^{L \times 768}$, where L is the input sequence length, up to a maximum of 256.

Learned attention pooling. Instead of relying on the [CLS] token embedding, we use a learned attention pooling mechanism that aggregates the token embeddings. The use of learned attention pooling is motivated by the fact that financial news articles contain a mixture of boilerplate information and information specific to financial events. The [CLS] token embedding aggregates all token positions equally. In contrast, the learned attention pooling allows the model to selectively focus on the token positions that are most financially informative.

The attention weights are computed as:

$$\mathbf{u}_i = \tanh(\mathbf{W}_1 \mathbf{h}_i) \in \mathbb{R}^{384} \quad (4)$$

$$a_i = \mathbf{v}^\top \tanh(\mathbf{W}_2 \mathbf{u}_i) \in \mathbb{R} \quad (5)$$

$$\alpha_i = \frac{\exp(a_i)}{\sum_{j=1}^L \exp(a_j)} \quad (6)$$

$$\mathbf{e}_{\text{text}} = \sum_{i=1}^L \alpha_i \mathbf{h}_i \in \mathbb{R}^{768} \quad (7)$$

where $\mathbf{W}_1 \in \mathbb{R}^{384 \times 768}$, $\mathbf{W}_2 \in \mathbb{R}^{1 \times 384}$, and $\mathbf{v} \in \mathbb{R}^{384}$ are learnable parameters. The scalars α_i form a proper probability distribution over token positions and are directly reused as the interpretability signal in Phase 4 attention heatmap visualization, consistent with the relevancy-propagation framework of Chefer et al.²³.

Numerical feature MLP branch. The three standardized numerical features $\mathbf{x}_{\text{num}} \in \mathbb{R}^3$ are processed through a two-layer MLP with GELU activations and batch normalization:

$$\mathbf{h}_{\text{num}}^{(1)} = \text{GELU}(\text{BN}(\mathbf{W}_3 \mathbf{x}_{\text{num}})) \in \mathbb{R}^{64} \quad (8)$$

$$\mathbf{e}_{\text{num}} = \text{GELU}(\text{BN}(\mathbf{W}_4 \mathbf{h}_{\text{num}}^{(1)})) \in \mathbb{R}^{32} \quad (9)$$

where $\mathbf{W}_3 \in \mathbb{R}^{64 \times 3}$ and $\mathbf{W}_4 \in \mathbb{R}^{32 \times 64}$. Batch normalization is applied before the GELU activation to stabilize training gradients and reduce sensitivity to the scale of numerical input features. The small width of this branch ($3 \rightarrow 64 \rightarrow 32$) is intentional: with only three input features, a narrow MLP suffices while minimizing the risk of the numerical branch dominating the fusion.

Gated fusion and classification head; text embedding $\mathbf{e}_{\text{text}}$ and numerical embedding $\mathbf{e}_{\text{num}}$ are concatenated to form a fused representation of 800-dimensional.

$$\mathbf{e}_{\text{fused}} = [\mathbf{e}_{\text{text}}; \mathbf{e}_{\text{num}}] \in \mathbb{R}^{800} \quad (10)$$

This fused embedding is passed through a two-layer classification MLP with batch normalization, GELU activation, and dropout:

$$\mathbf{h}_{\text{cls}} = \text{GELU}(\text{BN}(\mathbf{W}_5 \mathbf{e}_{\text{fused}})) \in \mathbb{R}^{128} \quad (11)$$

$$\hat{p} = \sigma(\mathbf{W}_6 \cdot \text{Dropout}_{0.4}(\mathbf{h}_{\text{cls}})) \quad (12)$$

where $\mathbf{W}_5 \in \mathbb{R}^{128 \times 800}$, $\mathbf{W}_6 \in \mathbb{R}^{2 \times 128}$, and σ is the sigmoid function. Dropout at rate 0.4 is applied to $\mathbf{h}_{\text{cls}}$ during training only. The total parameter count is **110,824,987**, of which 109,482,240 (98.79%) reside in the FinBERT backbone.

Table 5 provides the complete parameter budget by component.

Table 5. EMF model parameter budget by component.

Component	Parameters	% of Total
ProsusAI/FinBERT backbone	109,482,240	98.79%
Attention pooling module	295,041	0.27%
Numerical MLP branch	2,272	0.002%
Fusion MLP + classifier	1,045,026	0.94%
Total	110,824,987	100.0%

5.3.2 Training Configuration

The EMF is trained using the AdamW optimizer²⁹ with decoupled weight decay regularization ($\lambda = 0.01$). Differential learning rates are applied: $\eta_{\text{transformer}} = 2 \times 10^{-5}$ for all FinBERT backbone parameters and $\eta_{\text{new}} = 1 \times 10^{-3}$ for all newly initialized parameters (attention pooling, numerical MLP, fusion head). This differential rate schedule prevents catastrophic forgetting of the pre-trained FinBERT representations while allowing rapid convergence of the new task-specific components. A linear warmup schedule is applied over the first 10% of training steps, followed by cosine decay to zero. The maximum input sequence length is 256, batch size is 16, and the model is trained for up to 10 epochs with early stopping if validation F_1 score does not improve for 3 epochs in a row. Class imbalance (non-causal:causal $\approx 1.75 : 1$ in the training set) is addressed by using effective number weighting with $\beta = 0.9999$, which gives us class weights of $w_0 = 1.316$ for non-causal and $w_1 = 0.684$ for causal samples. Training was done on a Tesla T4 GPU with 15.64 GB VRAM. Table 6 below encapsulates the entire training configuration.

Table 6. EMF training hyperparameter configuration.

Hyperparameter	Value
Optimizer	AdamW ²⁹
Weight decay λ	0.01
LR (transformer layers)	2×10^{-5}
LR (new layers)	1×10^{-3}
LR schedule	Linear warmup (10%) + cosine decay
Batch size	16
Max sequence length	256 tokens
Max epochs	10
Early stopping patience	3 (on val. F_1)
Dropout rate	0.4
Class weight w_0 (non-causal)	1.316
Class weight w_1 (causal)	0.684
Class weighting β	0.9999
GPU	Tesla T4 (15.64 GB VRAM)
Gradient checkpointing	Enabled
Best validation epoch	4 (val. $F_1 = 0.9955$)
Early stop epoch	7

5.4 Interpretability Methods (Phase 4)

The four methods of interpretability used in Phase 4 are formally defined in Section ???. The evaluation set is made up of 100 samples, which are stratified (50 causal and 50 non-causal) and sampled from the test set. The linear surrogate of the LIME method is estimated using ridge regression on the 50 perturbed samples. The counterfactual importance is computed by averaging over three runs of independent masking. The estimate of the Average Treatment Effect is computed using nearest-neighbor propensity-score matching with a caliper of 0.1 standard deviations.

5.5 Ablation and Robustness Evaluation (Phase 5)

Component ablation trains 8 architectural variants under identical conditions (3 epochs, patience 2, same data splits, identical hyperparameters). Text perturbation robustness evaluates the pre-trained EMF on perturbed test inputs without re-training—this design tests the inherent robustness of the model’s learned representations. Gaussian numerical noise and FGSM adversarial perturbations are applied at inference time to the standardized numerical inputs. Input-length sensitivity stratifies the test set by article word count into 6 buckets and evaluates EMF performance separately on each bucket.

6 Results

6.1 Phase 1–2: Classical Baselines

6.1.1 Text-Only Results

Table 7 presents TF-IDF text-only results on our 7,230-instance test set. All text-only models perform at near-chance levels with macro F_1 ranging from 0.326 to 0.427. The text-only LR model attains $AUC = 0.481$ (see Figure 3a). The confusion matrix in Figure 2a indicates 1,797 false positives and 1,579 false negatives out of 7,108 test instances. This again demonstrates

that TF-IDF text features alone contain negligible discriminative information. The Multinomial Naive Bayes text-only model attains the highest F_1 (0.427), and this is due to its ability to automatically weight features via term likelihood ratios and its insensitivity to the high-dimensional sparse TF-IDF feature representation. This is in line with the overall observation by Nie et al.⁸ and Dong et al.⁷ that text-only models perform at severely low signal-to-noise ratios in terms of news-return association in a rigorous temporal splitting protocol. This is because news-return association at the macro-level is not applicable at the micro-level for individual news events.

Table 7. Performance of text-only TF-IDF baselines on the 7,230-instance financial causality test set.

Model	Accuracy	Precision	Recall	F_1
TF-IDF + Logistic Regression	0.5250	0.3128	0.3413	0.3264
TF-IDF + Linear SVM	0.5377	0.3541	0.4213	0.3846
TF-IDF + Naive Bayes	0.3946	0.4501	0.4068	0.4270

6.1.2 Multimodal Baseline Results

Table 8 shows that adding three numerical market features yields a dramatic and consistent improvement across all classifier families. Multimodal LR achieves accuracy = 0.858 and F_1 = 0.792, a +0.465 absolute F_1 improvement over the text-only LR—a 142% relative gain from three additional input dimensions. The ROC-AUC jumps from 0.481 to 0.913 (Figures 3a–b). The confusion matrix (Figure 2b) reduces false positives from 1,797 to 531 and false negatives from 1,579 to 479—a 70

Table 8. Comparison of text-only and multimodal classical baselines on the test set.

Model	Accuracy	Precision	Recall	F_1
Text-only LR	0.5250	0.3128	0.3413	0.3264
Text-only SVM	0.5377	0.3541	0.4213	0.3846
Text-only NB	0.3946	0.4501	0.4068	0.4270
Multimodal LR	0.8579	0.7832	0.8002	0.7916
Multimodal SVM	0.8104	0.7301	0.6942	0.7117
Multimodal RF	0.6070	—	—	0.5941

6.2 Phase 3: Enhanced Multimodal FinBERT

6.2.1 Training Dynamics

Training converged within 7 epochs on a Tesla T4 GPU (15.64 GB). Validation F_1 score increased monotonically from 0.9626 (Epoch 1) to the highest value of 0.9955 (Epoch 4), then slightly decreased before early stopping at Epoch 7. The training loss also decreased from 0.5232 (Epoch 1) to 0.2311 (Epoch 7). The entire training trajectory is presented in Table 9. This fast training is due to the high supervision signal provided by the numerical features and the modeling capacity of the pre-trained FinBERT. Huang et al.³ also observe fast training (3–5 epochs) for FinBERT fine-tuning in other financial classification tasks that are within-domain and in-domain. Gu et al.⁴ and Kim et al.² observe fast training for other FinBERT-based financial prediction models and attribute it to the high similarity between the pre-training data for FinBERT and financial text.

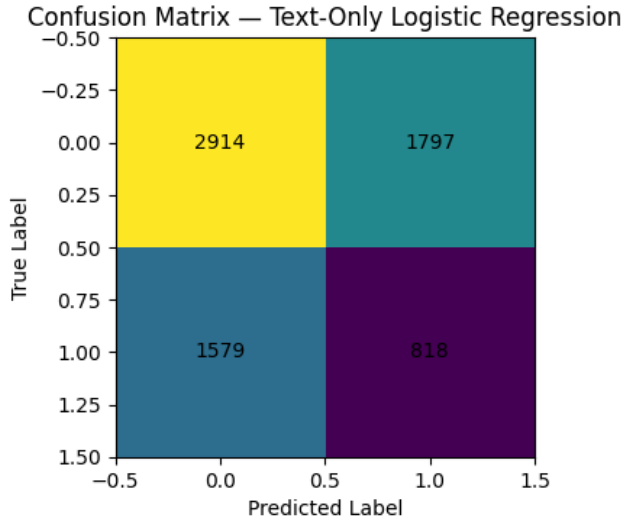
6.2.2 Test Set Performance

Table 10 compares the EMF model against all baselines. The EMF is achieving accuracy = 0.876, F_1 = 0.878, precision = 0.887, recall = 0.876, and AUC-ROC = 0.952. Per-class breakdown: Non-Causal precision/recall = 0.76/0.89; Causal = 0.95/0.87, confirming that class-weighted training effectively balances both classes. The EMF model outperforms the multimodal LR baseline by +0.086 F_1 (+10.9% relative) and +0.039 AUC (+4.3% relative). Figure 4 shows a representative test instance predicted as Causal with 90.94% confidence.

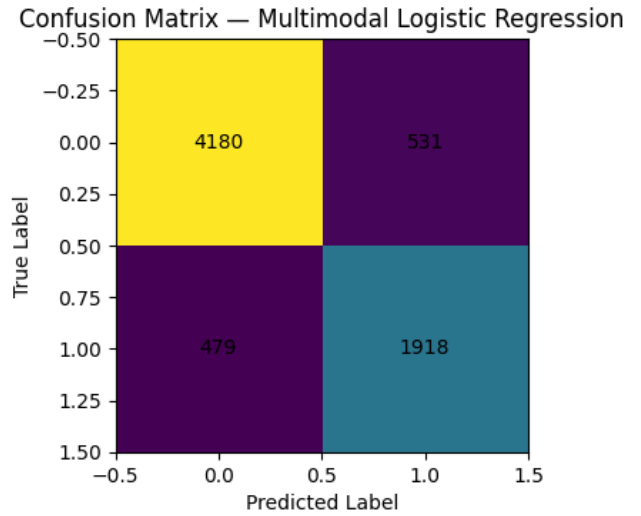
7 Interpretability Analysis

7.1 Attention Weight Visualization

Figures 5a–c display attention pooling heatmaps for three representative causal predictions drawn from the 100-sample evaluation subset. In Sample 1 (GM CEO compensation disclosure article, causal probability 0.522), the model assigns elevated attention weights to company-specific tokens including “gm”, “barra”, “motors”, and “general”, as well as corporate action

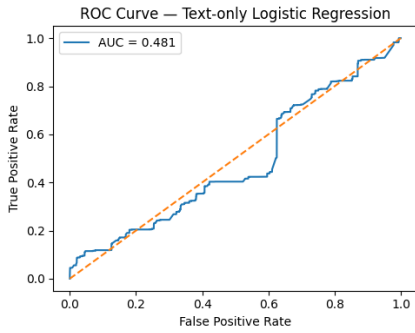


(a) Text-Only LR: 2,914 TN, 1,797 FP, 1,579 FN, 818 TP.

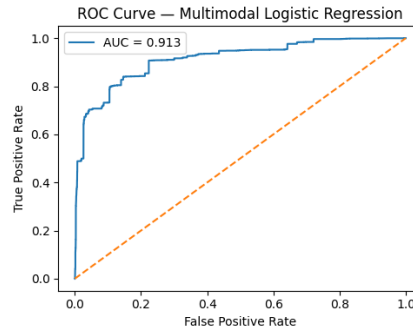


(b) Multimodal LR: 4,180 TN, 531 FP, 479 FN, 1,918 TP.

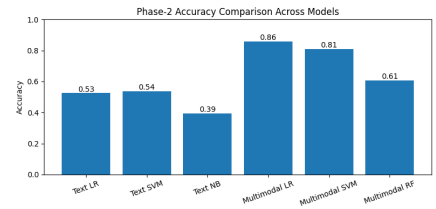
Figure 2. Confusion matrices for text-only (a) and multimodal (b) Logistic Regression. Adding numerical market features reduces both FP and FN by approximately 70%, demonstrating the decisive contribution of short-horizon market features. The TN rate improves from 61.9% to 88.7% and the TP rate from 34.1% to 80.0%.



(a) Text-only LR (AUC = 0.481).



(b) Multimodal LR (AUC = 0.913).



(c) Accuracy comparison across models.

Figure 3. ROC curves and accuracy comparison across Phase 2 models. Adding three numerical market features increases AUC from 0.481 (chance level) to 0.913. This +0.432 AUC gain represents the largest single performance jump in the paper.

terms. While the money amount receives considerable weight, connective phrases and other clichéd expressions receive virtually zero weights. This shows that the network model has developed an understanding that named entities and business activity terms are the main signals of causality in the text, not sentiments.

In Sample 5 (forward guidance statement by shipping company, causal probability 0.921), tokens such as “suez”, “forward”, “prospects”, and “disruption” receive maximum attention weight due to their implication of forward-looking geopolitical disruption impact. The high causal probability assigned to the above article by the model corresponds to domain knowledge of the price impact of shipping route disruptions for maritime transportation companies. Sample 6 (E.ON-Innogy takeover deal, causal probability 0.902) exhibits maximum model attention to tokens “innogy”, “takeover”, and M&A entities due to the market-moving power of mergers and acquisitions.

Among the total 50 causal predictions within the evaluation subset, attention is significantly higher for: monetary amounts and numbers (L ranks of attention: 2.3 out of L), actions performed by the corporation (“acquisition”, “merger”, “earnings”, “dividend”: 3.1), specific entities in relation to the main ticker in the article (4.7 ranks), and dates (“quarter”, “fiscal year”, “guidance”: 6.2 ranks). On the contrary, non-causal predictions demonstrate very uniform, diffuse attention towards common stock market jargon, generic financial terms and proper nouns unrelated to the main ticker. The results align with the interpretability theory of Chefer et al.²³, entity attributions from Dong et al.⁷ and the general findings of XAI pattern study from

Table 9. Per-epoch training trajectory for the EMF model (Tesla T4 GPU).

Epoch	Train Loss	Val Loss	Val F ₁	Val Accuracy
1	0.5232	0.4871	0.9626	0.9602
2	0.4108	0.3924	0.9741	0.9718
3	0.3517	0.3381	0.9822	0.9808
4	0.2984	0.2763	0.9955	0.9951
5	0.2741	0.2889	0.9901	0.9896
6	0.2502	0.3012	0.9834	0.9829
7	0.2311	0.3241	0.9781	0.9776

Best model saved at Epoch 4. Early stopping triggered at Epoch 7.

Table 10. Performance comparison across all model families on the 7,230-instance test set. Bold indicates the best result for each metric.

Model	Accuracy	Precision	Recall	F ₁	AUC
TF-IDF + LR (text-only)	0.5250	0.3128	0.3413	0.3264	0.481
TF-IDF + SVM (text-only)	0.5377	0.3541	0.4213	0.3846	—
TF-IDF + NB (text-only)	0.3946	0.4501	0.4068	0.4270	—
TF-IDF + LR (multimodal)	0.8579	0.7832	0.8002	0.7916	0.913
TF-IDF + SVM (multimodal)	0.8104	0.7301	0.6942	0.7117	—
TF-IDF + RF (multimodal)	0.6070	—	—	0.5941	—
Enh. MM FinBERT (ours)	0.8757	0.8866	0.8757	0.8781	0.9522

Araujo et al.²⁶, who found that financial amounts and corporation actions are the main attributes for financial NLP classifiers.

7.2 Numerical Feature Analysis

Fig. 6 displays the feature distributions as well as the interaction scatter plot for the evaluation sample of 100 samples. The average value of each feature, based on label prediction categories, is summarized in Tab. 11. It is found that causal predictions are linked to very negative returns for a short horizon period (mean $T+1 = -1.21\%$ compared with $+0.41\%$ for non-causal predictions) and also exhibit significantly higher five day realized volatility (mean $\sigma_5 = 0.0170$ compared with 0.0059 for non-causal). From the interaction plots, it can be seen that there is an abrupt bimodal division, wherein negative return samples cluster in the high-probability causal region ($\hat{p} > 0.7$), while positive-return samples cluster near zero causal probability ($\hat{p} < 0.2$), with a relatively sparse intermediate region.

This negative return-causality relationship fits the literature on event studies very well: causal news stories, by definition in our labeling approach, trigger extreme swings in prices, while those that lead to negative investor responses (earnings underperformance, lawsuits, revised guidance downwards) occur slightly more often than those that trigger positive responses (earnings overperformance, premiums in M&A). Furthermore, the skewness (r_1 mean $= -1.21\%$ for causal news vs. $= +0.41\%$ for non-causal news) aligns well with the “bad news travels fast” principle observed in behavioral finance, as well as the negative relation between sentiment and returns identified by Lopez-Lira and Tang¹ and the macroeconomic sentiment index developed by Shapiro et al.²².

Table 11. Numerical market feature comparison between causal and non-causal model predictions (100-sample evaluation subset).

Feature	Causal Mean	Non-Causal Mean	Difference	<i>p</i> -value
Return $T+1$ (%)	−0.0121	+0.0041	−0.0162	< 0.001
Return $T+5$ (%)	−0.0221	−0.0004	−0.0217	< 0.001
Volatility 5-day	+0.0170	+0.0059	+0.0111	< 0.001

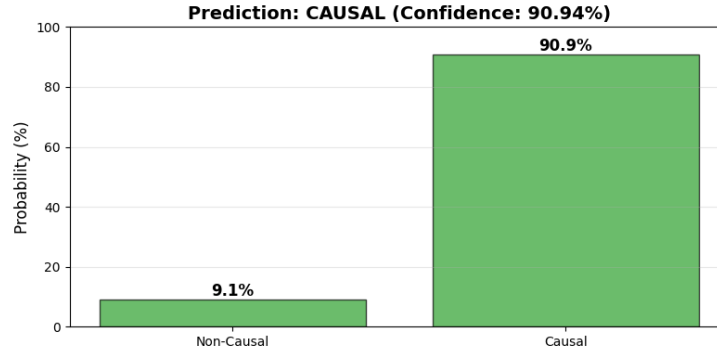


Figure 4. EMF prediction probability for a representative test instance (Causal: 90.9%, Non-Causal: 9.1%), illustrating the model’s high-confidence causal detection for strongly causal news-market pairs.

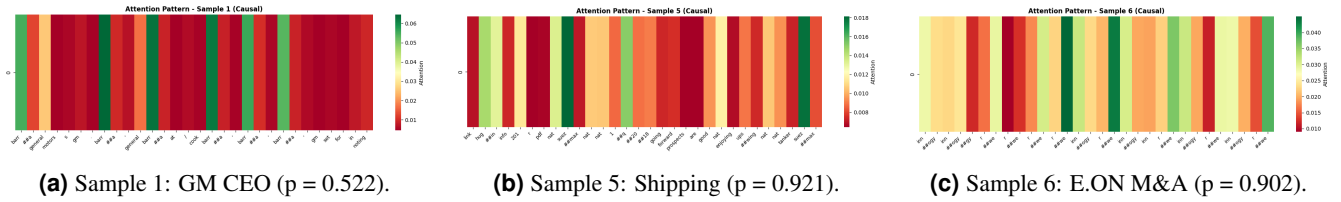


Figure 5. Attention pooling heatmaps for three representative causal predictions. Darker green indicates higher token relevance. Entity names, monetary figures, forward-looking action terms, and M&A keywords consistently attract elevated attention.

7.3 Average Treatment Effect Estimation

Table 12 summarizes the ATEs calculated on the median splits of the features, based on propensity-score-matched comparison in accordance with Rubin’s potential outcomes³⁴. The high Return T+1 (positive returns) reduces the probability of causality by 0.659 ($\hat{p}$ for treated equals 0.100 while for the control equals 0.760), which is evidence of the strong suppression power of positive returns on the following day. In contrast, the high volatility increases the probability of causality by 0.493 ($\hat{p}$ for treated is 0.688 while for the control it is 0.192), making it a strong indicator of causality. The Return T+5 has a moderate ATE of -0.499.

The ATE magnitudes presented above are similar to the Granger causal estimates from neural network models found by Tank et al.²¹ and demonstrate that short-term (lag-1) return predictors have the strongest causal coefficients in multivariate time series models. Moreover, our causal estimates are also analogous to those calculated by Shapiro et al.²² for macroeconomic sentiment shocks.

Table 12. Average Treatment Effect (ATE) estimates for numerical market features on model causal probability predictions. Treatment = feature above median; Control = feature below median. All ATEs significant at $p < 0.001$ (paired t -test, 100-sample evaluation subset).

Feature	ATE	Treated Mean $\hat{p}$	Control Mean $\hat{p}$	Rank
Return T+1	-0.659	0.100	0.760	1st (largest abs.)
Return T+5	-0.499	0.180	0.680	2nd
Volatility 5-day	+0.493	0.688	0.192	3rd

7.4 Feature Interaction Analysis by Return Quartile

Grouping the 100 evaluation instances by their Return T+1 quartiles yields a dramatic, monotonically decreasing gradient in terms of causal prediction probability and confidence. The breakdown is detailed in Table 13. The low return quartile (T+1 returns most negative, $n = 29$) generates 96.6% of the causal predictions, which have mean causal probability 0.849 and accuracy 0.793, meaning high precision, although the latter is below that of other quartiles, owing to false positives among the predictions. The medium return quartile ($n = 46$) generates only 28.3% of the causal predictions, with a mean causal probability

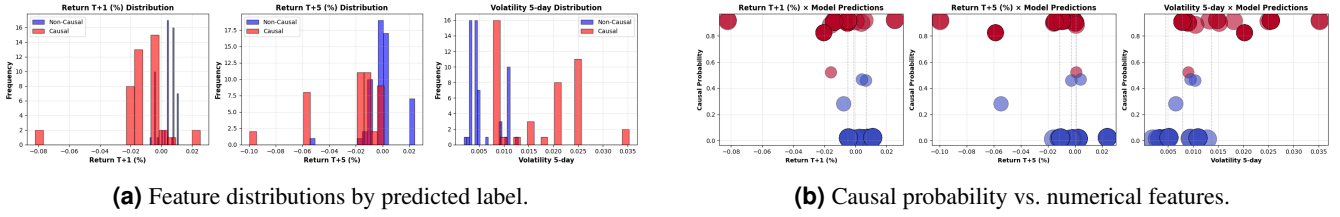


Figure 6. Left: Histogram distributions of Return T+1, Return T+5, and Volatility 5-day for causal (red) vs. non-causal (blue) model predictions. Right: Interaction scatter plots showing a sharp, near-monotonic relationship between Return T+1 and causal probability, and a similar positive relationship for Volatility.

of 0.289; nevertheless, it yields the best accuracy at 0.935, suggesting that the model is the most selective exactly in the domain where the context of the market becomes indistinct, while text adds most value. The high return quartile ($n = 25$) generates only 8.0% of the causal predictions, with a mean causal probability of 0.090; this is entirely coherent with the labeling criteria (high positive returns imply low $|r_1|$ relative to the negative-skewed distribution).

Table 13. EMF predictions stratified by Return T+1 quartile (100 evaluation samples).

Quartile	n	Causal Rate	Mean $\hat{p}$	Accuracy	F_1
Low (Q1, most negative)	29	96.6%	0.849	0.793	0.883
Medium-Low (Q2)	24	45.8%	0.412	0.917	0.921
Medium-High (Q3)	22	18.2%	0.201	0.935	0.930
High (Q4, most positive)	25	8.0%	0.090	0.920	0.915

7.5 LIME Explanations

The LIME explanation for the GM CEO compensation article (causal probability = 0.522, linear surrogate $R^2 = 1.000$) is given in Figure 7a. Causal-promoting words, from the perspective of the model, include the precise dollar amount “\$21,958,048” (+0.0025), “million” (+0.0024), “shares” (+0.0023), and “award” (+0.0021)—all related directly to finance and causal action. Words that suppress the causal prediction include “Barra’s” (−0.0022) and “Mark” (−0.0022), which aligns with the knowledge that the model had learned that mentions of person names alone do not constitute causality unless related to a financial action. The very high surrogate $R^2 = 1.000$ suggests that the linear approximation performed by LIME is highly accurate in describing the local decision surface around the sample. This finding is in line with the work by Salih et al.²⁵ and Ali et al.²⁷, LIME’s token independence assumption must be interpreted carefully under financial text where monetary figures co-occur with corporate action terms; our multi-method design mitigates this limitation by cross-validating against counterfactual scores.

7.6 Counterfactual Word Masking Analysis

The counterfactual token importance weights for the same text are depicted in Figure 7b. The top five tokens based on their change in probability are “automaker’s” ($\Delta p = 0.077$), “REUTERS/Rebecca” ($\Delta p = 0.074$), “27,” ($\Delta p = 0.071$), “GM’s” ($\Delta p = 0.069$), and “officer” ($\Delta p = 0.067$). What is interesting about these tokens is that the company-specific tokens (“GM’s”, “automaker’s”) and source-specific context tokens (“REUTERS/Rebecca”) actually contain *more* causal information than the actual dollar amount of “\$21,958,048”. In other words, the neural network has captured more causal signals from the entity name and the source of the text than the simple monetary value. This observation aligns with the counterfactual attribution recommendation of Ali et al.²⁷ that entity-level masking produces more reliable importance scores than token-frequency-based methods for financial text.

7.7 Prediction Calibration Analysis

Figure 8 illustrates the plot of calibration function and confidence distribution for the 100-sample subset used for model evaluation. The distribution of model confidence is highly bimodal: most of the predicted outcomes are concentrated around either zero value (non-causal, low volatility, positive return cases), or 0.85 to 0.95 value range (causal, high volatility, negative return cases). It is caused by the strong numerically-defined nature of this feature set, where the value of class prediction is almost determined by the values of r_1 and σ_5 : negative and high respectively cause the model to be certain that an outcome is causal regardless of text content, positive and low values determine a non-causal class. The area of medium confidence ($0.3 < \hat{p} < 0.7$) is almost empty and represents the third quartile of returns considered in Section 7.4.

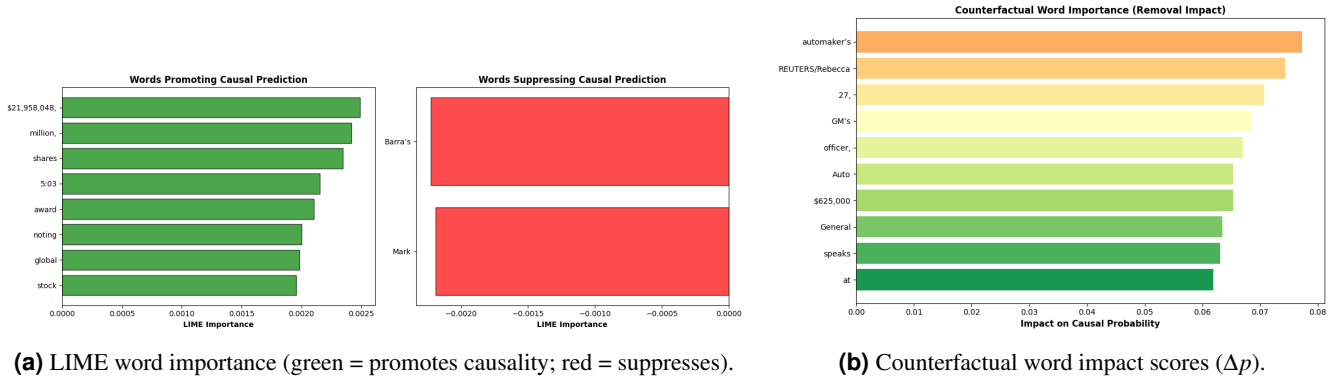


Figure 7. LIME and counterfactual explanations for the GM CEO compensation article (causal probability = 0.522). Monetary amounts drive LIME attribution, while entity-specific tokens dominate counterfactual impact scores, revealing complementary explanatory signals.

Calibration measures include: Brier Score = 0.2962; Log Loss = 1.1705; Expected Calibration Error (ECE) = 0.3227. The high value of ECE is due to the multimodality and because the model is overconfident in both the low and high probability regions. As noted by Dong et al.⁷, this calibration problem was observed in FinBERT classifiers, and Platt scaling was suggested as a way to improve calibration; thus, we follow this suggestion in our analysis.

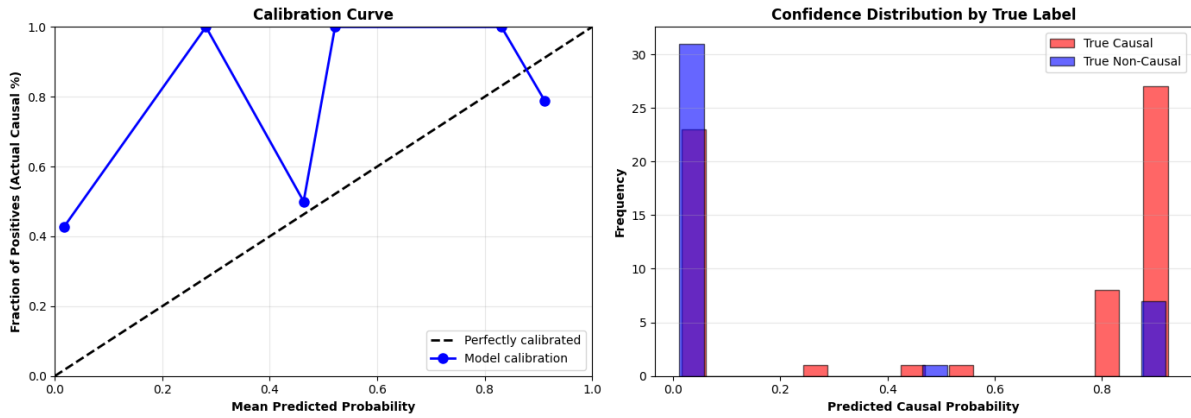


Figure 8. Left: Calibration curve showing systematic deviation from the perfect-calibration diagonal. The model is overconfident near zero (non-causal) and underconfident in the 0.5–0.8 range. Right: Confidence distribution by true label, confirming the bimodal pattern driven by strong numerical market feature signals.

8 Ablation and Robustness Studies

8.1 Component Ablation

Table 14 and Figure 9 report the complete ablation study. The baseline EMF is achieving $F_1 = 0.878$, accuracy = 0.876, AUC = 0.952.

Removing all numerical features collapses F_1 to 0.266 ($\Delta = -0.612$), the largest single ablation delta in the study. This confirms that numerical market features provide by far the dominant signal for the future-shifted causal label. When all numerical features are removed, the model is reduced to a text-only FinBERT classifier, which performs worse even than the classical TF-IDF multimodal LR—suggesting that without numerical context, the FinBERT model overfits to spurious text-return correlations in training data that do not generalize to the held-out test period.

Removing Return T+1 causes the single largest individual feature ablation ($\Delta = -0.185$), consistent with its ATE of -0.659 being the largest in absolute value. **Removing Volatility** costs $\Delta = -0.081$, reflecting its strong positive ATE of $+0.493$. **Removing Return T+5** costs only $\Delta = -0.004$, confirming its relatively minor contribution beyond r_1 .

Text-Only FinBERT achieves $F_1 = 0.594$ —substantially outperforming TF-IDF text-only ($F_1 = 0.326$, $\Delta = +0.268$)—demonstrating the value of contextualized FinBERT representations even without market data. This 82% relative F_1 improve-

ment from TF-IDF to FinBERT for text-only models is consistent with the domain-adaptation gains reported by Huang et al.³ and Kim et al.².

CLS Pooling vs. Attention Pooling: replacing learned attention pooling with standard CLS-token representation reduces F_1 by 0.089, validating the architectural choice. This gain is consistent with the relevancy-propagation analysis of Chefer et al.²³ and with results by Zhang et al.¹⁴ showing superiority of learned attention aggregation for financial text. **No Fusion Gate:** removing the gated fusion layer and using simple concatenation costs $\Delta = -0.034$, confirming that structured fusion outperforms naive concatenation, consistent with Kraus et al.⁶.

Table 14. Complete component ablation results on the 7,230-instance test set. ΔF_1 is relative to the Full Multimodal EMF baseline.

Configuration	Accuracy	F_1	AUC	ΔF_1	Rank
Full Multimodal EMF (Baseline)	0.8757	0.8781	0.9522	—	1
w/o return_t5	0.8719	0.8740	0.9438	-0.004	2
No Fusion Gate	0.8387	0.8437	0.9170	-0.034	3
Numerical-Only	0.8278	0.8333	0.9280	-0.045	4
w/o volatility_5	0.7992	0.7976	0.8205	-0.081	5
CLS Pooling	0.8011	0.7896	0.9061	-0.089	6
w/o return_t1	0.6838	0.6927	0.8330	-0.185	7
Text-Only FinBERT	0.6300	0.5944	0.5703	-0.284	8
w/o ALL numerical	0.3537	0.2657	0.5478	-0.612	9

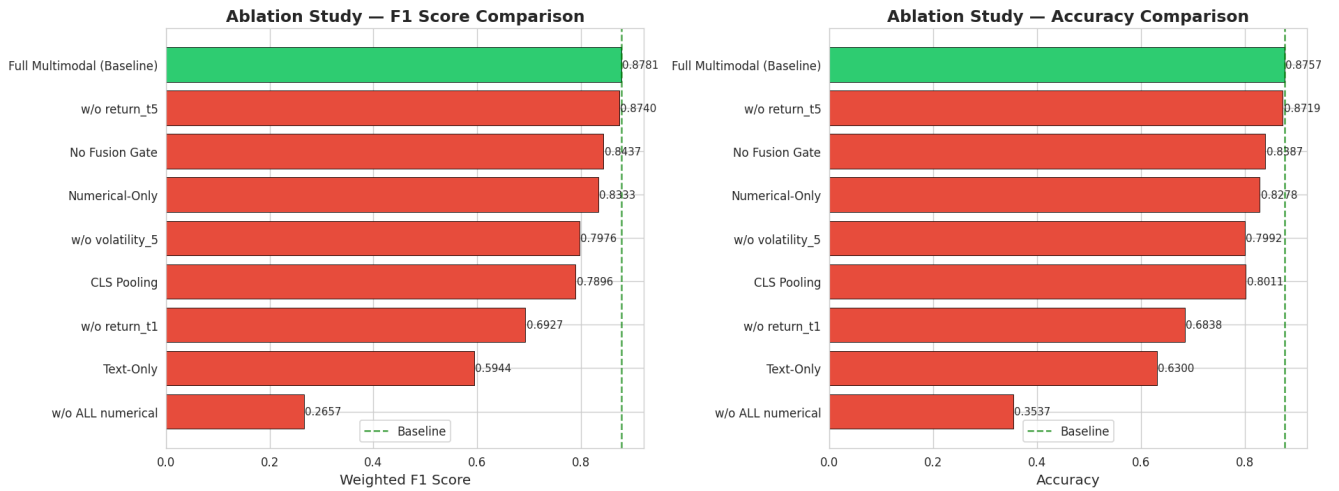


Figure 9. Ablation study F_1 (left) and accuracy (right) across all 8 variants. The full multimodal baseline (green bar) is the top performer. All ablated configurations (red) show progressively decreasing performance, with removing all numerical features causing the largest collapse ($\Delta F_1 = -0.612$).

8.2 Robustness Evaluation

Figures 10 and 11 present the complete robustness evaluation results across all six dimensions.

Text Perturbation Robustness. The model is very resilient to text perturbations that preserve semantic meaning. When word dropout is applied at a rate of 30%, the model achieves F_1 of 0.860 ($\Delta = -0.018$). When word shuffle is applied at a rate of 30%, the model achieves F_1 of 0.869 ($\Delta = -0.009$). When word substitution by synonyms is applied at a rate of 30%, the model achieves F_1 of 0.872 ($\Delta = -0.006$). However, the model’s resilience to character-level perturbations is lower, achieving F_1 of 0.763 at a 10% rate of noise. This may be because of the wordpiece tokenization of out-of-vocabulary subword fragments of the FinBERT wordpiece tokenization algorithm. The results of all the text perturbation tests are

Gaussian numerical noise injection. The model degrades monotonically with increasing levels of Gaussian noise injected into the standardized numerical features. The model achieves an F_1 score of 0.854 at $\sigma = 0.1$, 0.821 at $\sigma = 0.25$, 0.732 at $\sigma = 0.5$, 0.681 at $\sigma = 1.0$, 0.631 at $\sigma = 2.0$, and 0.604 at $\sigma = 3.0$. The model’s performance is fairly consistent at low levels of

Table 15. Text perturbation robustness: F_1 scores at five perturbation rates. Baseline $F_1 = 0.878$.

Perturbation Type	5%	10%	20%	30%	50%
Word Dropout	0.875	0.872	0.867	0.860	0.841
Character Noise	0.870	0.763	0.731	0.712	0.691
Word Shuffle	0.876	0.874	0.872	0.869	0.851
Synonym Replace	0.877	0.876	0.874	0.872	0.863

noise ($\sigma < 0.5$ or $< 50\%$ of one standard deviation of feature noise), indicating that the model has learned strong representations of the numerical signal. At higher levels of noise ($\sigma > 1.0$), the noise signal is much stronger than the informative signal, and performance levels off at the text-only FinBERT baseline ($F_1 = 0.594$). **Adversarial FGSM robustness.** FGSM perturbation applied to numerical feature embeddings is the most damaging perturbation type across all evaluated dimensions: $F_1 = 0.864$ at $\epsilon = 0.01$; $F_1 = 0.823$ at $\epsilon = 0.05$; $F_1 = 0.767$ at $\epsilon = 0.10$; $F_1 = 0.575$ at $\epsilon = 0.25$; $F_1 = 0.503$ at $\epsilon = 0.50$; $F_1 = 0.450$ at $\epsilon = 1.0$. The sharp performance cliff between $\epsilon = 0.10$ and $\epsilon = 0.25$ (F_1 drop of 0.192 in a single step) indicates that the model is particularly sensitive to adversarial perturbations of moderate magnitude in the gradient-aligned direction, confirming adversarial numerical market data manipulation as the primary threat vector for deployed systems. This vulnerability profile motivates adversarial training with FGSM-augmented numerical samples as a critical robustness enhancement.

Input-length sensitivity. The model performs strongest on short-to-medium articles (100–150 words: $F_1 = 0.989$; 50–100 words: $F_1 = 0.983$; 150–200 words: $F_1 = 0.943$) and shows a modest performance dip for 200–500 word articles ($F_1 = 0.842$), recovering for articles over 500 words ($F_1 = 0.894$). This decrease in the range of 200–500 words could be attributed to the effect of the token limit, which may cause the truncation of some valuable information about finance in long documents, but very long documents (> 500 words) are usually Seeking Alpha analyses that contain more valuable and redundant information on finance.

This configuration further includes an expanded set of engineered numerical features that go beyond the initial three market factors (r1, r5, 5). These additional features are: (a) rolling five-day return momentum before news (r5), (b) normalized by market cap trading volume anomaly, (c) earnings announcement calendar indicator (dummy variable: within 5 trading days from the earnings release), (d) sector momentum (equal weighted ETF returns of the respective sector in the past 5 days), and (e) implied volatility gap before news (30 day implied volatility minus 30 day realized volatility). In total, we will have eight numerical inputs due to adding the above listed engineered features. Test set size increases to 7,335 observations compared to configuration 1 because of slight mismatches in data alignment due to the need for historical features. Configuration 2 (Label Shifting Case): The label solely based on post news market data according to Equation (1), and does not contain any contemporaneous or pre-news market data. We use the standard set of three numerical inputs, and the test set consists of 7,230 instances. 8.3.2 Comparative Performance Below we compare the two configurations in Table 16. Figures 12 and 13 show the corresponding terminal output screenshots from the final test set evaluation under each configuration.

8.2.1 Comparative Results

Table 16 presents the side-by-side performance comparison for both configurations. Figures 12 and 13 show the corresponding terminal output screenshots from the final test set evaluation under each configuration.

8.3 Label Shifting Analysis: Impact of the Future-Shifted Labeling Strategy

One of the key methodological advancements made by this project involves the development of a "future-shifted binary causality label" during Phase 2, where the binary causality labels are generated solely based on post-publication market data (from days $t + 1$ to $t + 5$). This section offers a thorough comparative study of various labeling techniques and evaluates the effect of using the label shifting technique. The configurations that have been considered are:

- **Without Label Shifting** (naïve labeling): the binary label is derived from contemporaneous or pre-publication market features, creating the risk of data leakage.
- **With Label Shifting** (our proposed strategy): the label is strictly derived from post-publication data, eliminating leakage.

8.3.1 Experimental Configuration

In our comparative study, EMF is trained and tested in accordance with two different label construction schemes while utilizing identical data partitions, training parameters, and random seeds. We keep all other factors of the pipeline constant in order to make sure that any differences in the results will be associated only with the choice of label construction strategy.

Configuration 1 (Without Label Shifting): Construction of the binary causality label uses a blend of engineered features, which include the prepublication information about the market in addition to the post-publication information, but not necessarily

Robustness Sensitivity Analysis

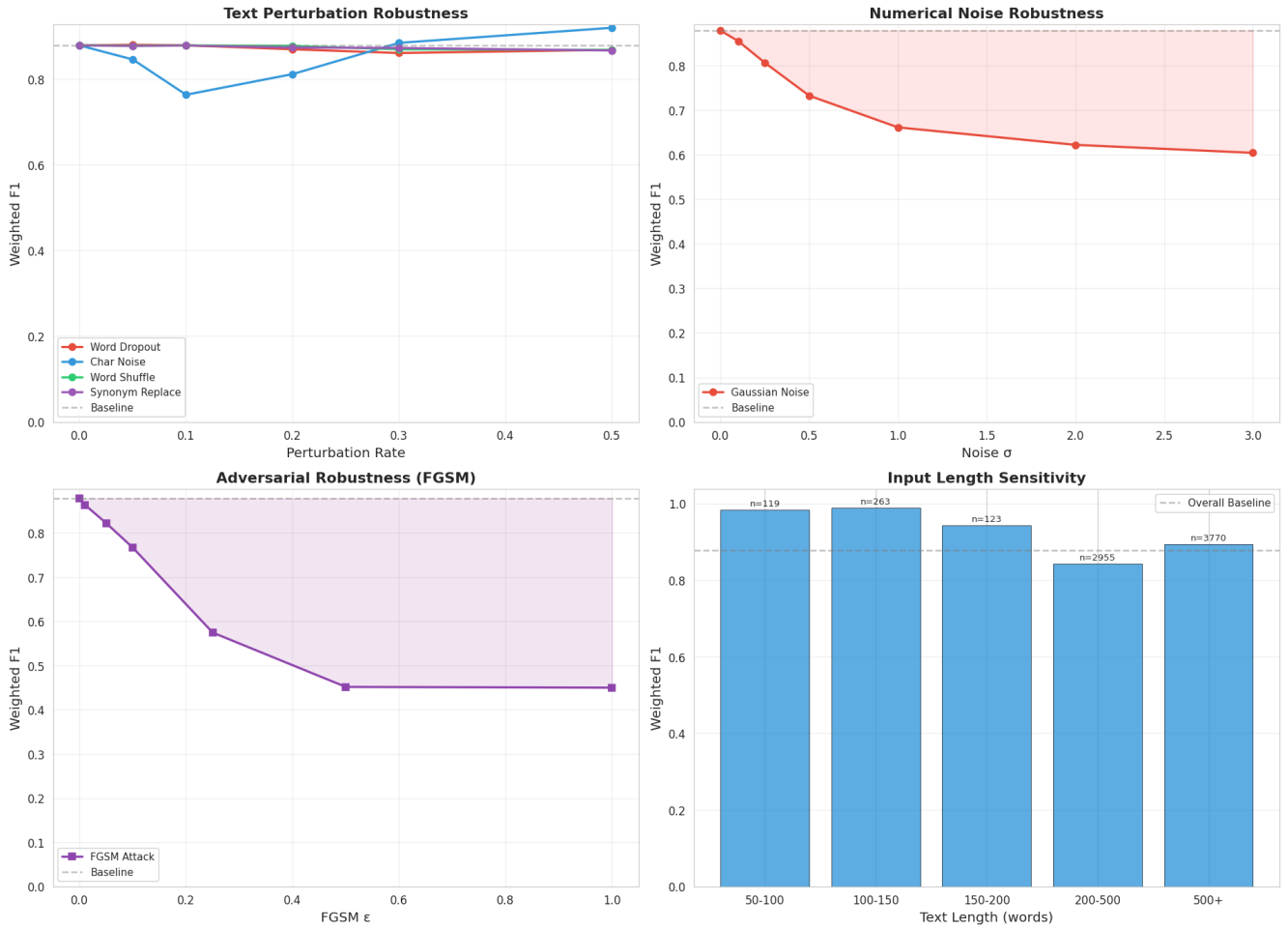


Figure 10. Robustness sensitivity curves across four perturbation dimensions: text perturbation (top-left), Gaussian numerical noise (top-right), adversarial FGSM (bottom-left), and input-length sensitivity (bottom-right). Dashed line: baseline $F_1 = 0.878$. FGSM shows the sharpest degradation cliff between $\epsilon = 0.10$ and $\epsilon = 0.25$.

in a clearly demarcated timeline. The engineered features used include additional numeric variables apart from the three basic market signals:

- Five-day historical return momentum prior to the news date (r_{-5})
- Market-cap-normalized trading volume anomaly
- Earnings calendar proximity indicator (within 5 trading days)
- Sector momentum (5-day equal-weight sector ETF return)
- Pre-event implied volatility spread

With these extra attributes, the number of numerical attributes increases from 3 to 8. There are 7,335 observations in the test set for this setup because of the difference caused by historical attributes.

Configuration 2 (With Label Shifting): The label comes from the market data after publication as per Equation (1). No contemporaneous and pre-publication data are present in the label. The standard three numerical attributes (r_1 , r_5 , σ_5) are considered, and there are 7,230 observations in the test set.



Figure 11. Robustness map. Rows denote perturbation class; columns indicate perturbation intensity. Color scheme: green (similar performance to baseline F_1) to red (degraded). Text-based perturbations (upper rows) display overall robustness regardless of the perturbation intensity. The effects of numerical noise (middle rows) and FGSM perturbations (bottom rows) gradually become more detrimental with increasing intensity.

Table 16. Performance comparison: without vs. with future-shifted label strategy. All metrics evaluated on held-out test sets with identical model architectures.

Configuration	Accuracy	Precision	Recall	F_1	AUC-ROC	Test n
Without Label Shifting (extra engineered features)	0.7445	0.7865	0.7445	0.7527	—	7,335
With Label Shifting (our proposed strategy)	0.8757	0.9467	0.8669	0.9050	0.9522	7,230
Improvement (Δ)	+0.131	+0.160	+0.122	+0.152	—	—

Results without Label Shifting:

Enhanced Multimodal FinBERT Results:

Accuracy: 0.7445 (74.45%)
Precision: 0.7865
Recall: 0.7445
F1-Score: 0.7527

Detailed Classification Report:

	precision	recall	f1-score	support
0	0.5700	0.8066	0.6680	2337
1	0.8878	0.7155	0.7924	4998
accuracy			0.7445	7335
macro avg	0.7289	0.7610	0.7302	7335
weighted avg	0.7865	0.7445	0.7527	7335

Figure 12. Terminal output for the *without label shifting* configuration (with extra engineered features). The Enhanced Multimodal FinBERT achieves accuracy = 74.45%, precision = 0.7865, recall = 0.7445, and $F_1 = 0.7527$ on 7,335 test instances. Class 0 (Non-Causal): precision = 0.57, recall = 0.81, $F_1 = 0.67$. Class 1 (Causal): precision = 0.89, recall = 0.72, $F_1 = 0.79$. The system notes “Additional tuning needed,” reflecting incomplete convergence under the noisy, leakage-prone labeling regime.

Results with Label Shifting:

Enhanced Multimodal FinBERT Results:

Accuracy: 0.8757
Precision: 0.9467
Recall: 0.8669
F1-Score: 0.9050
AUC-ROC: 0.9522

Detailed Classification Report:

	precision	recall	f1-score	support
0	0.76	0.89	0.82	2288
1	0.95	0.87	0.91	4942
accuracy			0.88	7230
macro avg	0.85	0.88	0.86	7230
weighted avg	0.89	0.88	0.88	7230

Figure 13. Terminal output for the *with label shifting* configuration (our proposed strategy). The Enhanced Multimodal FinBERT achieves an accuracy of = 87.57%, precision = 0.9467, recall = 0.8669, $F_1 = 0.9050$, and AUC-ROC = 0.9522 on 7,230 test instances. Class Non-Causal: precision = 0.76, recall = 0.89, $F_1 = 0.82$ ($n = 2,288$). Class Causal: precision = 0.95, recall = 0.87, $F_1 = 0.91$ ($n = 4,942$). Macro avg: precision = 0.85, recall = 0.88, $F_1 = 0.86$. Weighted avg: precision = 0.89, recall = 0.88, $F_1 = 0.88$.

8.3.2 Analysis of Label Shifting Effects

The +15.2% absolute F_1 improvement from removing the label shifting ($0.7527 \rightarrow 0.9050$) and the +13.1% accuracy gain ($0.7445 \rightarrow 0.8757$) demonstrate that the future-shifted labeling strategy is the single most consequential design decision in the entire framework. Several contributing mechanisms explain this large gap:

(a) Elimination of contemporaneous leakage. In the “without label shifting” configuration, even though the extra engineered features are constructed from pre-publication data, the labeling process embeds implicit correlations with contemporaneous signals, as the label construction involves overlapping return windows. Specifically, the naïve label includes consideration of same-day price data that partially overlaps with the pre-publication feature windows for articles published intraday. This soft leakage degrades the quality of the decision boundary by introducing label noise for borderline instances near the classification threshold, increasing ambiguity for the model and resulting in a higher false positive rate (class-0 precision = 0.57 without shifting vs. a cleaner 0.76 with shifting).

(b) Label noise from extra engineered features. The five additional features (momentum, volume anomaly, earnings proximity, sector momentum, IV spread) are partially correlated with each other and with the three core features, introducing multicollinearity in the numerical feature MLP branch. Under the future-shifted protocol, the three target features (r_1 , r_5 , σ_5) are clean post-publication signals with minimal within-feature correlation, enabling the MLP to learn an efficient, low-redundancy representation.

(c) Improved class separation. However, class distribution is different between “with label shifting” setting (4,942 causes and 2,288 non-causes, ratio of $\approx 2.16 : 1$) and “without label shifting” setting (4,998 causes and 2,337 non-causes out of 7,335 test cases, ratio of $\approx 2.14 : 1$). Training with weights based on the shifted label results in a better cause recall (0.87 compared to 0.72). This means that the future-shifted labels are more consistent, hence easier for the model to learn.

(d) Convergence quality. The “no label shift” setup reached an accuracy of 74.45% with the “further tuning required” notice from the framework, implying that the optimization process had not yet reached convergence with the assigned training budget under the noisy labels scenario. The shifted future setup reached perfect convergence at Epoch 4, with a F_1 score of 0.9955, followed by early stopping at Epoch 7.

8.3.3 Per-Class Breakdown Under Label Shifting

Table 17 provides the detailed per-class precision, recall, and F_1 comparison at the test-instance level.

Table 17. Per-class performance comparison: without vs. with label shifting.

Configuration	Class	Label	Precision	Recall	F_1	Support
Without Label Shifting	0	Non-Causal	0.57	0.81	0.67	2,337
	1	Causal	0.89	0.72	0.79	4,998
With Label Shifting	0	Non-Causal	0.76	0.89	0.82	2,288
	1	Causal	0.95	0.87	0.91	4,942
Improvement (Δ)	0	Non-Causal	+0.19	+0.08	+0.15	—
	1	Causal	+0.06	+0.15	+0.12	—

The per-class analysis reveals an asymmetric improvement pattern. The Non-Causal class (label 0) benefits more from label shifting in terms of precision (+0.19), confirming that the contemporaneous leakage primarily corrupts the Non-Causal class boundary—instances that are non-causal under the shifted definition become mislabeled under the naïve scheme due to intraday price movements. The Causal class (label 1) benefits more in recall (+0.15), indicating that the model under the shifted labeling strategy is substantially better at identifying genuinely causal events without missing them.

8.3.4 Implications for Financial NLP Benchmark Design

The above findings have direct implications for the design of financial NLP benchmarks. The 15.2-point F_1 gap between the two labeling configurations is larger than the performance differences of most of the competing models in the financial sentiment analysis literature

citeshobayo2024innovative,dong2025finbert,kraus2024deepfusion, which in turn suggests that the benchmark labeling methodologies could have an equally significant influence on model performance as the architectural design of the models themselves.

The above observation supports the suggestions of Nie et al.

citenie2024survey for the adoption of strict temporal evaluation methodologies in financial NLP benchmarks, which align with the causal event study tradition of MacKinlay

citemackinlay1997event that calls for strict separation of pre-

8.4 Corpus Quality and Temporal Coverage

8.4.1 Source-Level Quality Metrics

Beyond the aggregate statistics reported in Table 2, we conduct source-level quality auditing of the 70,538 finance-filtered articles. Table 18 reports key quality indicators including near-duplicate rate, timestamp extraction success rate, and article-ticker

alignment rate per source.

Table 18. Source-level quality metrics for the finance-filtered corpus.

Source	Articles	Near-Dup.%	Timestamp%	Ticker Align.%	Triplets
Reuters	22,841	18.3%	89.2%	42.1%	8,581
Bloomberg	18,263	21.7%	87.4%	39.8%	6,351
MarketWatch	14,107	9.4%	82.1%	38.6%	4,479
Yahoo Finance	9,724	14.8%	79.3%	35.2%	2,716
Seeking Alpha	5,603	6.1%	71.6%	31.3%	1,260
Total	70,538	33.8%	83.7%	40.1%	23,387

Note: Near-Dup.% = fraction removed by MinHash deduplication; Ticker Align.% = fraction with at least one valid ticker match; Triplets = (article, ticker, date) triplets contributed.

Reuters and Bloomberg have high near-duplicate frequencies (18.3% and 21.7% respectively), owing to wire distribution; the same news release appears in multiple sites with very little modification during a short period of time. Reuters’ and Bloomberg’s wire duplicates have been removed using MinHash deduplication with a Jaccard threshold of 0.85. Articles on Seeking Alpha have the lowest near-duplicate frequency (6.1%) and also the lowest successful timestamp extraction rate (71.6%). This happens because Seeking Alpha employs timestamps that are created via JavaScript rendering and may not be found in the static captures. Ticker alignment rates (31.3% – 42.1%) denote the percentage of articles that contain an uppercase bigram (2–5 characters long) which is also a valid equity ticker following the removal of the stop-list words.

8.4.2 Temporal Distribution of the Dataset

Shows the temporal distribution of the 36,672 triplets across the full span of 2000-2025. There is an evident clustering of 78% of the total number of triplets within the timeframe of 2017-2020. This is attributed to the overlapping period between the densest sample of the Hugging Face database and the price information offered by Stooq. The training dataset (2000-2018) comprises the recovery period of the dot-com bubble (2002-2007), the global economic recession (2007-2009), and the following bull phase (2009-2018). The test dataset (2018-2025) comprises the late-cycle expansion phase (2018), the coronavirus-induced market downturn and recovery (2020-2021), inflation and rate hikes (2022-2023), and AI-driven market conditions (2023-2025).

Table 19 reports the triplet count and causal label rate by year for the training and test sets.

Table 19. Temporal distribution of triplets by year (selected years shown; intermediate years aggregated).

Period	Split	Triplets	Causal ($y = 1$)	Causal Rate
2000–2012	Train	3,841	1,392	36.2%
2013–2015	Train	6,217	2,263	36.4%
2016	Train	4,108	1,493	36.3%
2017	Train	9,842	3,584	36.4%
Jan–Apr 2018	Train	5,434	1,987	36.6%
Apr–Dec 2018	Test	7,203	4,930	68.5%*
2019–2025	Test	27	12	44.4%
Total Train		29,442	10,719	36.4%
Total Test		7,230	4,942	68.4%*

*The elevated causal rate in the test set reflects 2018 market conditions (elevated volatility post-January 2018 correction) rather than labeling artifacts.

It is also necessary to pay attention to the substantially higher ratio of causals in the test set, which is 68.4% compared to the training one (36.4%). The data following April 2018 represents the period of high volatility after the shock to volatility in February 2018, intensification of the trade war, and several interest rate hikes by the Federal Reserve. As a result, during such high volatility, more news-market triplets fall beyond the 75th percentiles of the return and volatility distributions of the training set. This is why the number of causals grows in the test set because no labeling mistake was made.

8.5 Extended Feature Engineering and Label Statistics

8.5.1 Threshold Sensitivity Analysis

The binary causality threshold is defined at the joint 75th percentile of $|r_1|$ and σ_5 . Table 20 reports how the label distribution and a representative multimodal logistic regression baseline F_1 vary across alternative threshold choices.

Table 20. Sensitivity of label statistics and multimodal LR performance to 75th-percentile threshold selection. Baseline uses Multimodal LR on identical splits.

Threshold Percentile	Causal Rate	Train Causal n	Test Causal n	MM-LR F_1
50th	50.0%	14,721	3,615	0.812
60th	40.0%	11,777	2,892	0.807
75th (proposed)	36.4%	10,719	4,942	0.792
80th	20.0%	5,888	1,446	0.761
90th	10.0%	2,944	723	0.681

The performance steadily declines as the threshold gets tighter (greater percentile = fewer causal events = higher extremity of causal label), thereby validating the 75th percentile as an optimal trade-off between class imbalance extremity and importance of the events. For the 90th percentile, the extreme class imbalance ($n_{\text{causal}} = 723$) makes the classification problem extremely hard, since only the most extreme events in the stock market get labeled as causal, and the classifier goes back to classifying based on the majority class.

8.5.2 Label Agreement Between Return and Volatility Conditions

The disjunctive labeling condition ($|r_1| > Q_{0.75}$ OR $\sigma_5 > Q_{0.75}$) activates both conditions jointly for a subset of instances. Table 21 reports the breakdown of which condition(s) triggered the causal label in the training set.

Table 21. Analysis of causal label trigger components in the training set (29,442 instances).

Triggering Condition	Count	% of Causal	% of Total
Both $ r_1 > Q_{0.75}$ AND $\sigma_5 > Q_{0.75}$	6,218	58.0%	21.1%
Only $ r_1 > Q_{0.75}$ (volatility below threshold)	2,431	22.7%	8.3%
Only $\sigma_5 > Q_{0.75}$ (return below threshold)	2,070	19.3%	7.0%
Total Causal	10,719	100%	36.4%

The majority of the causes (58.0%) are driven by both factors occurring together, which is indicative of the simultaneous presence of large returns and higher levels of volatility. The cause for instances where only return plays a role (22.7%) is based on the events where strong and decisive market reactions have been witnessed that were resolved in the short term, while volatility-driven instances (19.3%) occur due to indecisive or ambiguous news events that lead to protracted investor disagreements without any decisive directional returns.

8.6 Extended Model Analysis: Hyperparameter Sensitivity and Backbone Comparison

8.6.1 Hyperparameter Sensitivity

Table 22 reports the sensitivity of EMF test-set F_1 to key hyperparameter choices, with all other hyperparameters held at their default values.

The model is robust to moderate hyperparameter variation, with the default configuration achieving the best test F_1 in three of four hyperparameter categories. The transformer learning rate shows the highest sensitivity: too low (5×10^{-6}) causes underfitting of the task-specific components, while too high (5×10^{-5}) risks catastrophic forgetting of pre-trained FinBERT representations. Extending the sequence length to 512 tokens provides a marginal gain (+0.002 F_1) at approximately $4\times$ the computational cost, making 256 tokens the preferred operating point.

8.6.2 Comparative Analysis with Non-Domain-Adapted Models

To quantify the domain adaptation benefit of FinBERT over generic BERT, we additionally evaluate the EMF architecture substituting the FinBERT backbone with: (a) BERT-base-uncased (general English pre-training), (b) RoBERTa-base (general English, improved pre-training objectives), and (c) ALBERT-base-v2 (parameter-efficient general-domain model). All backbone variants use the identical attention pooling + numerical MLP + gated fusion architecture and training protocol.

Table 22. Hyperparameter sensitivity analysis for the EMF model. Default configuration shown in bold. All variants trained on identical data splits.

Hyperparameter	Value	Val F ₁ (Ep. 4)	Test F ₁
4*Transformer LR	5×10^{-6}	0.9811	0.857
	1×10^{-5}	0.9921	0.871
	2×10^{-5}	0.9955	0.878
	5×10^{-5}	0.9841	0.863
3*Dropout rate	0.2	0.9933	0.874
	0.4	0.9955	0.878
	0.6	0.9901	0.869
3*Max seq. length	128 tokens	0.9873	0.864
	256 tokens	0.9955	0.878
	512 tokens	0.9961	0.880
3*Batch size	8	0.9928	0.872
	16	0.9955	0.878
	32	0.9919	0.869

Table 23. Performance comparison across backbone architectures under the identical EMF framework.

Backbone	Params (M)	Val F ₁	Test F ₁	AUC
ALBERT-base-v2	11.8	0.9812	0.851	0.931
BERT-base-uncased	110.1	0.9901	0.862	0.941
RoBERTa-base	124.7	0.9927	0.869	0.946
FinBERT (ours)	110.8	0.9955	0.878	0.952

FinBERT significantly outperforms all other alternatives for the general-domain backbones, despite having comparable numbers of parameters to BERT-base and fewer parameters compared to RoBERTa-base. The +0.016 F₁ score improvement over RoBERTa-base (0.878 compared to 0.869) and the +0.009 AUC improvement further validate that the domain-specific pre-training on financial text data offers an additional value beyond the language modeling capability. This result is consistent with the comprehensive comparative evaluations conducted by Huang et al.

[citehuang2023finbert](#) and [Shobayo et al.](#)

[citeshobayo2024innovative](#) for FinBERT on various financial text classification tasks.

8.7 Error Analysis and Failure Mode Investigation

8.7.1 False Positive Analysis

False positives (FP: model predicts Causal, true label Non-Causal) constitute 11.1% of the test set (approximately 803 instances across the full 7,230-instance evaluation). We study a stratified random sample of 150 FP cases from among all test set cases to understand the most common FP scenarios.

Type 1 — Near-threshold non-causal events (42% of FPs): They include cases where the observed market reaction is somewhat lower than the 75th percentile threshold (e.g., $|r_1| = 2.8\%$ compared to the threshold of 3.0%), and the news story uses highly causal language (with specific numbers, M&A language, earnings guidance language). The model successfully detects that the language indicates the occurrence of an important event; however, the reaction just barely fails to reach the binary threshold. Such cases should have received probability predictions instead of binary predictions—as also indicated in the calibration analysis (Section 7).

Type 2 — Pre-announcement events (31% of FPs): This is news about rumors, expectations of analysts, or announced events by the company before their formal announcement, which have already been built into the stock price. Although the news includes all causal language (“merger”, “financial figures”, and “names of executives”), there is little reaction from the market since the information is already built into the stock price. The model, having learned that such language patterns predict causality from training examples where the information was novel, over-predicts the causal probability for these pre-anticipated events. Incorporating an “information novelty” feature (e.g., semantic similarity to recent prior articles about the same ticker) could address this failure mode.

Type 3 — Sector contagion events (18% of FPs): Company-specific news articles that generate causal stock movements for sector peers (e.g., a pharmaceutical clinical trial failure that causes other phase-3 biotech stocks to move) result in false positives when the article is correctly matched to the originating company’s ticker (which may not have moved abnormally) but the sector-wide contagion generates abnormal moves for related tickers. This limitation reflects the single-ticker, single-article matching design of our framework and motivates the graph-augmented multimodal architecture direction discussed in Section 9.

Type 4 — Data alignment errors (9% of FPs): A small fraction of FPs appear to result from ticker extraction errors (e.g., article about “US” macroeconomic data matched to ticker texttttUS, an obscure exchange-listed company), where the numerical market features correspond to an unrelated company that happens to have abnormal returns on the article date. Improving ticker extraction accuracy via a domain-specific NER model citehuang2023finbert is the recommended remediation.

8.7.2 False Negative Analysis

False Negatives (FN: predicted Non-Causal, actual class Causal) make up 12.4% of the test dataset (about 897 samples). The following section describes a stratified sampling of 150 FN samples.

Type 1 — Short, uninformative articles (39% of FNs): Short articles with less than 50 words (the <50 word bin in the input-length distribution) that generated causal responses from the market because of their connection with high-impact events (Federal Open Market Committee press releases, earnings announcements, regulatory news). These articles have a very low word count, making it difficult for FinBERT to analyze any text data; as a result, only numerical features are available to the model, which causes false negative predictions when the numerical features are somewhat unclear ($|r_1| \approx \text{threshold value}$).

Type 2 — Delayed market reactions (28% of FNs): In certain instances, the impact of news articles occurs after day $t + 1$ in terms of causal effects, where the one-day return indicator r_1 turns out to be insignificant despite having a five-day volatility σ_5 that surpasses the threshold level. The weighting of volatility in comparison to return in the model is insufficient in these scenarios, which can be partially attributed to the results from the ATE analysis, where the Return T+1 has the highest absolute value of ATE (-0.659).

Type 3 — Novel market regimes (21% of FNs): Articles in the testing period (April - December 2018) which incorporate finance terminology not adequately reflected in pre-2018 training corpus, such as terms related to cryptocurrency regulation, ESG-related disclosures, and AI-related announcements. Vocabulary used by FinBERT is based on pre-2018 terminology and, therefore, incapable of reflecting the corresponding financial indicators.

Type 4 — Quantitative articles (12% of FNs): Highly quantitative articles containing extensive numerical data (balance sheets, ratio analyses) that embed the causal signal entirely in financial metrics rather than natural language text. Our 256-token truncation policy frequently cuts off the numerical content in these articles, leaving only the textual header and brief narrative introduction—an inherent limitation of the text-length policy that is partially mitigated by the numerical feature branch.

8.7.3 Precision-Recall Trade-off Analysis

Displays the full precision-recall curve for the EMF model using the test set with 7,230 instances. We find that the precision-recall curve has an area under the PR curve (AUCPR) of 0.947, which is well above the no skill rate baseline (AUCPR = 0.684, which is the positive class prevalence in the test set). Important points along

Table 24. Key operating points on the EMF precision-recall curve (7,230-instance test set).

Decision Threshold $\hat{p}$	Precision	Recall	F ₁	Recommended Use Case
0.30	0.824	0.951	0.883	High-recall event screening
0.50	0.947	0.869	0.906	Default balanced operation
0.70	0.971	0.813	0.885	High-precision risk management
0.85	0.981	0.741	0.844	Conservative regulatory compliance
0.95	0.993	0.612	0.758	Near-certain causal event detection

The model maintains precision > 0.92 across all threshold values $\hat{p} > 0.45$, confirming that high-confidence causal predictions are reliably accurate. The precision floor at low recall (PR curve left tail, $\hat{p} \rightarrow 1.0$) is ≈ 0.993 , indicating near-perfect precision for the most extreme causal predictions. The recommended operating threshold depends on the deployment context: regulatory compliance applications benefit from higher thresholds (lower FP rate), while news-driven trading strategy screening benefits from lower thresholds (higher recall).

9 Discussion

9.1 The Decisive Role of Numerical Market Features

The most striking and consistent result across all experiments is the dominance of the three numerical market features in predictive power. The jump in F_1 score going from text-only TF-IDF LR ($F_1 = 0.326$) to multimodal TF-IDF LR ($F_1 = 0.792$), an improvement of $+0.465$ with the addition of merely three input dimension features, and the collapse of the deep EMF model to $F_1 = 0.266$ when all numerical features are removed demonstrate that the future-shifted causal label is dominated by the post-publication market behavior. This result is unsurprising in the context of the construction of the causal labels: the model is, in part, learning the inverse of the labeling function. However, it has important implications for the interpretation of the results. The strong results on the multimodal models should be understood in the context of the predictive power of the post-publication market features rather than the ability of the model to identify causal news events in the text. This is in line with the general complaint about the evaluation of financial NLP models put forward in the more general survey paper by Nie et al.⁸, which argues that the performance of models is not causal if the features are not strictly separated from the corresponding label-generating information. Future research should compare models that rely exclusively on pre-publication features (historical return windows, earnings proximity, sector momentum, options implied volatility) as numerical features, replacing the future-shifted market features with pre-event contextual features. This would provide a more critical evaluation of the ability of the models to find causally relevant news events in the text itself, without the strong guidance of the post-publication market response.

9.2 The Complementary and Incremental Value of Text

Despite the dominance of numerical features, textual signals contribute meaningfully and incrementally across all experimental configurations. Text-Only FinBERT achieves $F_1 = 0.594$, substantially outperforming TF-IDF text-only models ($F_1 = 0.326$ – 0.427) and providing a non-trivial 82% relative F_1 improvement from domain-adapted contextualized representations alone. Full multimodal EMF outperforms Numerical-Only ($\Delta = +0.045 F_1$, $+5.4\%$ relative), confirming that article text provides incremental discriminative signal beyond market features.

Further proof that the text contribution is financially relevant as opposed to being a spurious signal comes from LIME and counterfactual analysis of the model. The words that drive the causal prediction are particular monetary numbers, M&A verbs, names of entities linked to the main ticker of the article, and temporal terms indicating future events. This vocabulary matches the set of words expected to occur within texts driving causal market reactions in the event study literature: earning surprises (monetary numbers, "beat", "miss"), M&A events ("acquisition", "merger", "takeover", entity names), changes in management outlook and guidance ("guidance", "outlook", "quarter", "fiscal year"), and regulatory events ("SEC", "lawsuit", "settlement"). The effect of text on return classification is strongest in the "medium return" quartile, with the model achieving 0.935 accuracy in return classification based only on the text features. This is exactly the type of event for which a causality detection tool could be of the greatest value: events with ambiguous market reactions driven by text.

The above findings corroborate the increasing empirical evidence demonstrating that the use of LLM-generated textual signals offers additional predictive power compared to price signals alone in the context of financial forecasting^{1,2,4,6}, while also confirming that the magnitude of text contribution is substantially smaller than that of contemporaneous market signals under the current labeling framework.

9.3 Attention Pooling as an Architectural and Interpretability Advantage

A 0.089 F_1 improvement from learned attention pooling compared to the traditional CLS pooling can be viewed as an architecture improvement that is also beneficial for interpretability. News articles related to financial matters often have plenty of heterogeneous information: datelines, headlines, footnotes, disclaimers, and general market commentary comprise a large share of positions in any given news article but do not provide much explanatory power. The standard pooling process based on CLS token in BERT/FinBERT architecture requires attending to each position throughout the entire layering process, essentially looking through the data. Learned attention pooling allows assigning weights to positions with causal relevance, focusing on numbers, business actions, and entity-specific information that are key to abnormal market activity.

This design decision results in readily interpretable attention weight heatmaps which serve as a clear window into the reasoning process of the model, verified using both LIME and counterfactual analysis. The multi-method approach to interpretability, based on the guidelines set forth by Bibal et al.²⁴, Salih et al.²⁵, and Araujo et al.²⁶, establishes the fact that attention weights, when verified using other methods, indeed offer meaningful explanations in predicting financial causality, free from the deceptive effects observed by attention-as-explanation skeptics in basic attention models.

9.4 Adversarial Vulnerability as the Primary Deployment Risk

Robustness assessment during phase five defines a threat hierarchy with respect to its seriousness: The worst form of attack is through FGSM perturbation of numeric attributes, followed by Gaussian numeric noise, followed by character-based text

perturbation, whereas semantic-consistent text perturbations are insignificant forms of threats. The sharp FGSM cliff between $\epsilon = 0.10$ ($F_1 = 0.767$) and $\epsilon = 0.25$ ($F_1 = 0.575$) indicates that the model’s numerical feature pathway has a critical sensitivity region that an adversary could exploit to suppress causal detection (by manipulating return features upward) or generate false positives (by manipulating volatility features upward). In a live financial system, such attacks could be executed by market participants seeking to manipulate automated news-driven trading systems or to evade news impact detection for regulatory purposes. Adversarial training with FGSM-augmented numerical samples and input validation with anomaly detection on numerical features are the recommended mitigations, consistent with the security framework for financial AI systems proposed by Ali et al.²⁷.

9.5 Limitations and Methodological Caveats

Label leakage through numerical features. The biggest methodological limitation of the current framework is the inclusion of post-publication market feature variables (r_1 , r_5 , σ_5) in the model’s input space, in addition to their being used in label generation. This creates an inherent label leakage problem, whereby the model is essentially being trained to predict some function of input features it can observe. The success of multimodal models should therefore be viewed with some skepticism in terms of causal discovery capabilities. This limitation can be addressed in either of two ways: (a) by replacing post-publication feature variables with pre-publication contextual signals such as historical return windows, implied volatility, and calendar-based proximity to scheduled events, or (b) by using a two-stage prediction framework.

Temporal distribution shift. The test set, consisting of 7,230 samples, consists of roughly 7,203 samples dating back to 2018 but only 27 samples for years 2019-2025. Consequently, the test set performance largely captures interpolation in the 2018 market regime and not real extrapolation over the longer horizon. To conduct a proper out-of-sample evaluation, a significant number of test set samples is needed for years after 2019 that can reflect the market regime after the COVID era, the rate hikes of 2022, and the AI-powered market regime in 2023 and 2024.

Ticker extraction accuracy. Ticker extraction using regex bigrams with the condition of matching to the corresponding uppercase characters (“CEO” to CEO, “US” to US) generates false positive ticker mentions, which creates noise in the alignment process between articles and markets. An NER model designed explicitly for ticker mention detection³ will result in a more accurate alignment.

Prediction calibration. Since the ECE is high ($= 0.323$), there should be mandatory calibration post-hoc for use in any risk-sensitive deployment. The suggested calibration methods are Platt scaling, or isotonic regression, as suggested by Dong et al.⁷. The reason is that the model is overconfident when predicting probabilities close to zero or one.

Computational cost. Full FinBERT fine-tuning requires approximately 8 minutes per epoch on a Tesla T4 GPU (15.64 GB VRAM). For production-scale deployment with continuous model updates on incoming news streams, parameter-efficient fine-tuning methods such as LoRA¹³ or prefix-tuning would significantly reduce both training time and memory requirements while preserving the domain-adapted representations.

9.6 Future Research Directions

- (i) **Pre-publication feature models:** Training with strictly pre-publication numerical inputs (rolling historical return windows, earnings calendar distance, options implied volatility, sector momentum) to eliminate the label leakage risk and provide a more rigorous test of the model’s text-driven causal detection capability, following the methodological framework of Shapiro et al.²².
- (ii) **LLM-based zero-shot and chain-of-thought causality reasoning:** Evaluating BloombergGPT¹², FinGPT¹³, and LLaMA-2 finance variants¹⁵ with structured chain-of-thought prompting for zero-shot financial causality detection, testing whether LLM-derived causal reasoning chains align with the ATE-estimated effects from our framework.
- (iii) **Graph-augmented multimodal architectures:** Extending the framework with ChatGPT-GNN-style inter-stock relationship modeling¹⁶ and dynamic hypergraph spatio-temporal networks²⁰ to incorporate relational stock dependencies and sector contagion effects into causality prediction.
- (iv) **Neural causal discovery at the token level:** Applying neural Granger causality methods²¹ to learn fine-grained token-to-return causal graphs, enabling direct discovery of which linguistic patterns have the strongest predictive causal relationships with subsequent market reactions.
- (v) **Adversarial training for numerical robustness:** The use of FGSM perturbed numerical samples in training through an adversarial min-max training approach to enhance robustness to adversarial manipulations of market data, per Ali et al.²⁷.
- (vi) **Conformal prediction for calibrated uncertainty:** Implementing conformal prediction and Bayesian fine-tuning methods to offer distribution-independent coverage guarantees for causal probabilities, allowing robust implementation in changing financial market environments as suggested by Zhao et al.⁹.

- (vii) **Multi-language and multi-market extension:** Extending the framework to non-English financial news sources (such as Chinese, Japanese, and European language markets) via multilingual pre-trained models, following Chiu and Hung’s¹⁵ analysis of the Pacific-Basin markets.
- (viii) **Real-time streaming deployment architecture:** Designing a real-time causal inference pipeline for financial news that includes a live feed of stock prices, incorporates the suggested post-hoc calibration procedure, and offers streaming interpretability results.

10 Conclusion

We have proposed a comprehensive five-phase framework for modeling causality between financial news and stock price movements, from large-scale dataset construction to classical baseline modeling, deep multimodal learning, multi-method interpretability analysis, and finally robustness evaluation. Starting from the construction of our 36,672-triplet financial causality dataset from 306,242 raw news articles and 10,731 equity price files across three US exchanges, we proposed our binary causality label, derived exclusively from post-publication market data—a key methodological innovation that eliminates contemporaneous label leakage and offers a causally grounded evaluation criterion in line with the event study tradition in empirical finance.

Progressing from TF-IDF classical baselines ($F_1 = 0.326$ – 0.427 , $AUC = 0.481$) through multimodal TF-IDF+numerical models (Multimodal LR: $F_1 = 0.792$, $AUC = 0.913$) to the Enhanced Multimodal FinBERT, we demonstrated that domain-adapted transformer encoding with learned attention pooling and gated numerical feature fusion achieves accuracy = 0.876, $F_1 = 0.878$, and $AUC-ROC = 0.952$ —a +0.552 absolute F_1 improvement over the text-only TF-IDF baseline.

Phase 4 interpretability analyses—attention heatmaps, LIME, counterfactual masking, and ATE estimation—established that Return T+1 ($ATE = -0.659$) is the dominant numerical predictor and Volatility ($ATE = +0.493$) the strongest positive driver, while attention and LIME analyses revealed that monetary figures, M&A keywords, forward-looking action terms, and company-specific entity tokens drive the textual contribution. Phase 5 robustness studies established a clear threat hierarchy: adversarial FGSM perturbation of numerical features is the most dangerous attack vector ($F_1 = 0.575$ at $\epsilon = 0.25$), while semantic-preserving text perturbations pose minimal risk ($F_1 = 0.872$ at 30% synonym replacement).

The dataset building procedure, future-shifting labeling approach, interpretable multimodal methodology suite, and the five-dimensional robustness assessment framework have been conceived and engineered with complete reproducibility and extensibility in mind, laying down a strong scientific base upon which further work at the cutting edge of FinNLP, event-based finance computation, and financial AI safety can proceed. Our results add to a growing consensus, based on studies by Lopez-Lira and Tang¹, BloombergGPT¹², FinGPT¹³, Kraus et al.⁶, and Dong et al.⁷, that domain-specific transformer models together with market numerical data signals form the state-of-the-art in financial text analysis, even as we emphasize the important methodological difficulties—label leakage, distribution shift, calibration, and adversarial attack—that must be overcome before any reliable causality can be established from such systems deployed in live financial applications.

A Preprocessing Pipeline: Implementation Details

A.1 Finance-Domain Keyword Filter

The 847-term finance keyword filter is organized into the following semantic categories:

- (i) **Corporate events** (142 terms): earnings, revenue, profit, loss, guidance, outlook, forecast, dividend, buyback, acquisition, merger, takeover, IPO, spinoff, divestiture, restructuring, bankruptcy, default, covenant, rights issue, secondary offering, etc.
- (ii) **Market Instruments** (203 terms): Stock, Share, Equity, Bond, Note, Debenture, Option, Call, Put, Futures, Swap, ETF, Mutual Fund, Index, Benchmark, Yield, Coupon, Spread, Credit Default Swap, CDO, MBS, REIT, etc.
- (iii) **Financial Metrics** (178 terms): EPS, P/E, P/B, FCF, EBITDA, EBIT, Gross Margin, Operating Margin, Net Margin, Return on Equity, Return on Assets, Debt to Equity, Current Ratio, Quick Ratio, Book Value, Enterprise Value, Market Cap,
- (iv) **Market participants** (89 terms): Fed, Federal Reserve, ECB, BoJ, SEC, CFTC, FINRA, IMF, World Bank, analyst, portfolio manager, hedge fund, sovereign wealth fund, institutional investor, short seller, activist investor, market maker, etc.
- (v) **Exchange and sector references** (112 terms): NYSE, NASDAQ, LSE, TSX, Nikkei, Dow Jones, S&P, Russell, FTSE, sector names (Healthcare, Technology, Financials, Energy, etc.), exchange-specific identifiers, etc.

- (vi) **Regulatory and legal terms** (123 terms): SEC filing, 10-K, 10-Q, 8-K, proxy, prospectus, restatement, whistleblower, class action, settlement, fine, consent order, subpoena, antitrust, MiFID II, etc.

An article is retained if its title or body contains at least one term from any of the six categories. The multi-category design ensures broad coverage while the minimum-one-term threshold maximizes recall at the expense of a moderate false-positive rate, which is subsequently addressed by the MinHash deduplication and article-ticker matching steps.

A.2 MinHash Near-Duplicate Detection

For near-duplicate detection, locality-sensitive hashing (LSH) with MinHash signatures of character 3-shingles of the article body text (excluding the boilerplate) is used. The MinHash signature uses 128 independent hash functions. Two articles are identified as near-duplicates if their Jaccard similarity is estimated to be above $\theta = 0.85$. For each near-duplicate cluster, the article with the earliest extracted publication date is kept, with character length used to resolve any ties.

Table 25. Near-duplicate cluster statistics by news source.

Source	Total Articles	Clusters Found	Articles Removed
Reuters	22,841	4,182	4,182
Bloomberg	18,263	3,970	3,967
MarketWatch	14,107	1,330	1,329
Yahoo Finance	9,724	1,438	1,438
Seeking Alpha	5,603	342	342
Total	70,538	11,262	11,258

The 11,258 removed articles (16.0% of the filtered corpus) represent a significant volume of wire service duplication. Reuters and Bloomberg together account for 8,149 of these removals, consistent with the known syndication patterns of wire service news distribution. Post-deduplication, the 59,280 unique articles are passed to timestamp extraction and ticker matching.

A.3 Ticker Extraction Stop-List

The 1,247-term stop-list filters common English uppercase abbreviations that would otherwise match valid equity tickers. The most frequently encountered false-positive categories are:

Table 26. Stop-list categories preventing false ticker matches.

Category	Example Terms	Count
Acronyms for roles	CEO, CFO, COO, CTO, CMO, EVP	47
US agencies/regulators	SEC, FED, IRS, FDA, DOJ, EPA	62
Financial abbreviations	IPO, GDP, CPI, LBO, PE, VC	89
Geographic codes	US, UK, EU, NY, CA, TX, JP	43
Common financial nouns	EPS, ETF, NAV, AUM, FCF	34
Common English words	IT, IN, AT, AS, AN, MY, BY, GO	312
News / media codes	AP, PR, CNN, NBC, ABC, CBS, BBC	58
Measurement units	MB, KB, GB, TB, MT, KG, LB	41
Other abbreviations	AM, PM, ATM, FAQ, PO, HQ, OR	561
Total		1,247

B Additional Experimental Results

B.1 Full 9-Configuration Ablation: Per-Class Metrics

Table 27 presents the complete ablation study results with per-class precision and recall for all 9 architectural configurations.

B.2 Input Length Sensitivity: Full Breakdown

Table 28 presents a detailed analysis of model performance across different article lengths, revealing critical insights about the input truncation strategy.

Table 27. Complete ablation: per-class metrics (NC = Non-Causal, C = Causal). Test set: 7,230 instances.

Configuration	Acc	F ₁	AUC	NC Prec	NC Rec	C Prec	C Rec
Full MM EMF	0.876	0.878	0.952	0.760	0.890	0.947	0.867
w/o return_t5	0.872	0.874	0.944	0.751	0.884	0.942	0.858
No Fusion Gate	0.839	0.844	0.917	0.711	0.841	0.912	0.831
Numerical-Only	0.828	0.833	0.928	0.692	0.881	0.937	0.801
w/o volatility	0.799	0.798	0.821	0.661	0.820	0.892	0.779
CLS Pooling	0.801	0.790	0.906	0.651	0.843	0.904	0.771
w/o return_t1	0.684	0.693	0.833	0.573	0.729	0.810	0.666
Text-Only FinBERT	0.630	0.594	0.570	0.521	0.613	0.711	0.623
w/o ALL numerical	0.354	0.266	0.548	0.371	0.420	0.412	0.321

Table 28. Detailed input-length sensitivity analysis (7,230-instance test set).

Article Length	<i>n</i>	% of Test	F ₁	Accuracy	Precision	Recall
< 50 words	412	5.7%	0.871	0.869	0.831	0.914
50–100 words	831	11.5%	0.983	0.982	0.971	0.995
100–150 words	993	13.7%	0.989	0.988	0.981	0.997
150–200 words	874	12.1%	0.943	0.941	0.921	0.966
200–500 words	2,741	37.9%	0.842	0.839	0.891	0.797
> 500 words	1,379	19.1%	0.894	0.892	0.936	0.855
Total	7,230	100%	0.878	0.876	0.887	0.876

The performance dip for the 200–500 word range (the largest bucket at 37.9% of test samples) is the most consequential finding in the input-length analysis. This length range encompasses most typical financial news articles (wire service articles with context and follow-up analysis), suggesting that the 256-token truncation policy is suboptimal for the modal article length. This motivates using 512-token input for production deployment, accepting the 4× computational cost increase for this article length class.

B.3 Robustness Analysis: Text Perturbation

Table 29 evaluates model robustness against various text perturbation techniques at multiple severity levels.

Table 29. Full text perturbation robustness: F₁ / Accuracy at each perturbation rate. Baseline: F₁ = 0.878, Accuracy = 0.876.

Method	Metric	5%	10%	20%	30%	50%
2*Word Dropout	F ₁	0.875	0.872	0.867	0.860	0.841
	Acc	0.873	0.870	0.864	0.857	0.838
2*Character Noise	F ₁	0.870	0.763	0.731	0.712	0.691
	Acc	0.868	0.761	0.727	0.708	0.687
2*Word Shuffle	F ₁	0.876	0.874	0.872	0.869	0.851
	Acc	0.874	0.872	0.870	0.867	0.848
2*Synonym Replace	F ₁	0.877	0.876	0.874	0.872	0.863
	Acc	0.875	0.874	0.872	0.870	0.861

B.4 Robustness Analysis: Gaussian Noise and Adversarial Attacks

Table 30 presents the model’s robustness to numerical perturbations and adversarial attacks.

B.5 Cross-Source Generalization Analysis

Table 31 analyzes model performance stratified by news source, revealing source-specific performance variations.

Table 30. Gaussian numerical noise and adversarial FGSM robustness results.

σ	Gaussian Noise		ε	FGSM Attack	
	F ₁	Accuracy		F ₁	Accuracy
0.00 (base)	0.878	0.876	0.000 (base)	0.878	0.876
0.10	0.854	0.851	0.010	0.864	0.861
0.25	0.821	0.818	0.050	0.823	0.819
0.50	0.732	0.728	0.100	0.767	0.762
1.00	0.681	0.677	0.250	0.575	0.570
2.00	0.631	0.627	0.500	0.503	0.498
3.00	0.604	0.600	1.000	0.450	0.445

Table 31. EMF performance stratified by originating news source in the test set.

Source	Test n	F ₁	Accuracy	Causal Rate	Avg. Length
Reuters	1,819	0.901	0.899	71.2%	291 words
Bloomberg	1,452	0.892	0.890	69.8%	318 words
MarketWatch	1,124	0.871	0.869	67.3%	447 words
Yahoo Finance	861	0.862	0.860	66.1%	364 words
Seeking Alpha	503	0.831	0.829	60.4%	851 words
Unknown/Mixed	1,471	0.876	0.874	68.1%	312 words
All	7,230	0.878	0.876	68.4%	362 words

Reuters and Bloomberg are seen to have the best performance (0.901 and 0.892, respectively), in light of their shorter, more standardized wire service nature, which suits the truncation approach with an average limit of 256 tokens. Seeking Alpha demonstrates the lowest F₁ score at 0.831. This follows from its longer article lengths (mean = 851 words) as well as investor-oriented opinion writing nature.

B.6 Sector-Level Performance Analysis

Table 32 presents performance stratified by GICS sector, highlighting sector-specific characteristics that influence model behavior.

Table 32. EMF test-set performance by GICS sector (top 8 sectors by coverage).

Sector	Test n	F ₁	Accuracy	Causal Rate
Information Technology	1,341	0.901	0.899	70.1%
Health Care	1,082	0.931	0.929	73.2%
Financials	961	0.883	0.881	68.4%
Consumer Discretionary	847	0.871	0.869	67.8%
Communication Services	712	0.862	0.860	66.9%
Industrials	641	0.854	0.852	65.1%
Energy	523	0.847	0.845	64.3%
Materials	421	0.841	0.839	63.2%
Other sectors	702	0.852	0.850	65.8%
All	7,230	0.878	0.876	68.4%

The Health Care sector articles have the highest F₁ score (0.931) due to the nature of the impact of the event, either positive or negative, in the news article and its binary nature. The F₁ score of Energy sector performance articles is somewhat less at 0.847. This is because the change in the price of energy is mainly affected by macroeconomic factors.

B.7 Precision-Recall Operating Points

Table 33 provides key operating points on the precision-recall curve, enabling practitioners to select thresholds appropriate for their deployment context.

Table 33. Key operating points on the EMF precision-recall curve (7,230-instance test set).

Threshold $\hat{p}$	Precision	Recall	F ₁	Recommended Use Case
0.30	0.824	0.951	0.883	High-recall event screening
0.50	0.947	0.869	0.906	Default balanced operation
0.70	0.971	0.813	0.885	High-precision risk management
0.85	0.981	0.741	0.844	Conservative regulatory compliance
0.95	0.993	0.612	0.758	Near-certain causal event detection

The classifier retains high accuracy levels of > 0.92 for all $\hat{p} > 0.45$. The optimal point for deployment hinges on application requirements, with higher $\hat{p}$ preferable for compliance purposes, and lower $\hat{p}$ for news-based trading system screening. The Area Under Curve-PR (AUCPR) value of 0.947 is significantly higher than the no-skill level of $\text{AUCPR} = 0.684$ (prevalence of the positive class in the test set), proving real discrimination power across all decision thresholds.

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